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Enhancing LLM Reasoning Capabilities Through Brokered Multi-Expert Reflection

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ABSTRACT Large Language Models (LLMs) have found increasing application in tasks requiring multi-step reasoning, yet challenges such as hallucinations and inconsistencies in the generated responses persist. This study presents an innovative methodology to enhance the reasoning capabilities of LLMs by brokering and integrating multiple expert LLMs within a reflection layer to provide targeted feedback on the reasoning trajectories of the base LLM. The approach employs a foundational pre-trained LLM as the base model, which is further supported by agents to promote cognitive assistance for specific task types. In instances where conclusions are deemed incorrect or reasoning is interrupted, these instances are forwarded to the expert LLM layer, which includes systems such as Claude-3 haiku for intricate contexts and MedAlpaca for medical reasoning, to deliver feedback on the base model's reasoning paths. This feedback forms a 'reflection pool,' enabling the base LLM to amend and enhance its reasoning trajectories in subsequent iterations. The experiments conducted across diverse datasets, including HotPotQA, SimpleQA, and PubmedQA, underscore the proposed architecture's efficacy in augmenting success signals, Rouge-L scores (indicative of quality and precision), and CTRLEval Consistency Scores (indicative of coherence and consistency). The architecture effectively addresses the issues of hallucinations and inconsistencies that frequently occur in multi-step reasoning. Importantly, the approach exhibits considerable potential in tackling domain-specific tasks, underscoring the importance of achieving correct and reliable conclusions. To facilitate further investigation and validation of our proposed brokered multi-expert reflection framework, we are open-sourcing the source code of our system, available at <https://github.com/WiZY936/Brokered-Multi-Expert-Reflection>.

INDEX TERMS LLM, self-reflection, chain-of-thought, multi-hop inference, feedback-driven refinement

I. INTRODUCTION

The advent of Large Language Models within the domain of artificial intelligence and its related systems has captivated the attention of both researchers and consumers. Large Language Models are extending the frontiers of artificial intelligence by transforming interactions between machines and humans, particularly in the spheres of automation, health-care, and real-time decision-making. Despite the promising opportunities presented by the development of LLM, such as intelligent automated systems, significant challenges remain that must be addressed.

LLM agents and brokers, especially when engaged in multi-step reasoning tasks, encounter difficulties with hallucination when attempting to sustain the context of analogous problems over extended durations, resulting in the generation

of irrelevant and erroneous responses [1]. Moreover, fine-tuning LLM agents for specific tasks remains challenging due to the scarcity of large, high-quality datasets and lack of extensive computational resources [2].

LLM prompt engineering has been proven to be an important skill to enhance the outputs of the LLMs [3]. This phenomenon naturally incites interest in experimenting with various prompting techniques to achieve the most optimal output from the LLM. Recent scholarly efforts focused on enhancing the task planning capabilities of LLM-based agents have systematically categorized them into Task Decomposition, Plan Selection, External Module, Reflection, and Memory. According to Xu *et al.* [4], despite Reflection consuming more tokens, the improvement in success rates for complicated tasks is promising, demonstrating the LLM's error-correcting

capability.

This study seeks to advance the development of more efficient methodologies for the creation and optimization of LLM-based agents capable of processing complex inputs provided to the LLM, building upon work disclosed in recent patents attributed to some of the authors. The research employs multiple expert fine-tuned LLMs to evaluate the errors of the base LLM within a multi-tier architecture that incorporates a layer dedicated to feedback-generating expert LLMs. Distinct sets of reflective LLMs are systematically selected in alignment with domain-specific tasks pertinent to the designated environment. The reflective data is preserved as Experience to facilitate the optimization of LLM-based agents for subsequent applications.

The motivation for this project largely stems from the study conducted by Huang, Xu, *et al.* [4], which shows that the extent of improved results of Self-Reflection based agents is much lower in Interactive Retrieval Environments (like HotPotQA [5] and FEVER [6]) and Interactive Programming Environments (like AgentBench [7] and MiniWoB++ [8]) compared to Interactive Action Environments (like ALF-World [9] and ScienceWorld [10]). Thus, a need for finer architecture is highlighted that observes better results across a broad spectrum of use cases.

II. RELATED WORK

Several studies provide a pertinent background for our work, which seeks to enhance the performance of LLMs using introspective, reflective, and reinforcement methodologies aimed at fulfilling goal-driven tasks. Concerning the mitigation of issues such as hallucinations in LLMs, which arise from ambiguous prompts and inadequate data quality, various studies have presented demonstrations, such as the study conducted by Ji, Ziwei, *et al.* [11] provides insights into performance advantages while introducing a self-reflection methodology. This study aims to reduce hallucinations, particularly in medical datasets, because of the significant risks hallucinations carry while dealing with such data. The proposed approach improves accuracy by utilizing an iterative self-reflection framework comprising three loops: (a) Factual Knowledge acquiring loop, where the LLM generates and refines the base knowledge until it reaches factual correctness; (b) Knowledge Consistent answering loop that verifies the consistency between generated knowledge and response; and (c) Question-Entailment answering loop which verifies if the final response logically aligns with the prompt. In terms of performance, this methodology provides promising results in addressing hallucinations specific to complex medical concepts in medical QA systems. Apart from this, the study investigates the proposed method solely on medical domain tasks, which should be further investigated to comprehensively understand multi-domain applicability of this approach.

With regard to improving the problem solving capabilities of LLM agents, there have been studies demonstrating that LLM agents can improve their problem solving abilities if provided with introspective capabilities. Qu, Yuxiao,

et al. [12] examine a framework (RISE) of using an interactive multi-round training on on-policy rollouts and supervision from a reward function, training models that are capable of improving themselves over multiple turns. For inference, the framework utilizes a majority voting system on outputs from different turns to obtain a brushed-up final response. There have been demonstrations suggesting that utilizing RISE significantly improves the self-improvement abilities of 7B models on reasoning tasks, while also being simpler, as it works well with just one model instead of running multiple models. Since RISE requires running manual iterations and improving with self-generated supervision, it holds a significant computational constraint, due to which it has only been tested for no more than two iterations. This restricts the scalability of on-policy learning in a data-efficient manner.

Another study examines the self-reflection method in multiple LLM agents [13], allowing them to generate better responses by reflecting on their Chain of Thought. The architecture outlined in this research consists of a base agent answering all the questions, after which the incorrect responses are queued for reflection. Self-reflecting agents then analyze the mistakes of the base agent and generate feedback with the help of correct answers already provided to them. For this, the answer labels and their descriptions are redacted to avoid leakage while generating the feedback. The base agent then utilizes these reflections to re-answer each incorrect question. The interpretation suggests that the reflections that contain more information outperform the reflections provided by agents that are restricted to provide only limited information. A drawback of this research is its reliance on using multiple choice questions for evaluation metrics, where the feedback agents are already provided with a correct answer label, which might not fully capture the inherent complexities of real-world reasoning scenarios. Finally, for all models and agents, the LSAT-AR (Analytical Reasoning) exam appears to be the most difficult and benefits from self-reflection. The significant performance jump from a single exam can skew the aggregate results across all exams, necessitating a set of exams with uniform difficulty to eliminate the skewness. In addition to these, Shengnan *et al.* [14] present a study that attempts to enhance the reasoning capabilities of LLMs on mathematical problem-solving tasks. The architecture of this study consists of generating mistake-correction data pairs (correction data) by employing multiple LLMs to generate inaccurate reasoning paths and using GPT-4 to generate corrections (identification, explanation, and guidance) for these paths. Next, the approach injects the correction data into the Chain-of-Thought data for the fine-tuning process to generate better reasoning paths. The authors find that the effectiveness of CoT data and correction data is non-homogeneous and a stronger backbone model can be more effective at learning from mistakes. While this approach can also facilitate specialized LLMs, the study suggests that less powerful models are not suitable for generating corrections, and even GPT-4 struggles to correct inaccurate reasoning paths for challenging questions.

Features	Our Approach	Ji, Ziwei, et al. [11]	Qu, Yuxiao, et al. [12]	Renze, et al. [13]	Shengnan et al. [14]	Zhang, et al [15]
Multi-Agent Framework	✓ (Multiple expert LLMs with dedicated feedback layer)	✗ (Single-agent reflection)	✗ (Single model with iterative improvement)	✓ (Multiple agents with reflection)	✗ (Single model with correction)	✓ (Personality-based collaborative agents)
Domain-Specific Expertise	✓ (Dedicated experts for medical, general reasoning)	✓ (Medical only)	✗ (General reasoning only)	✗ (General reasoning only)	✓ (Math only)	✗ (General reasoning only)
Chain-of-Thought Correction	✓ (Expert-driven refinement of reasoning trajectories)	✗ (Self-correction only)	✗ (Self-improvement only)	✓ (Peer reflection on CoT)	✓ (Expert-correction of math reasoning)	✗ (No explicit CoT correction)
Multi-Round Reflection	✓ (Structured iterative feedback with reflection pool)	✓ (Knowledge loops for consistency)	✓ (On-policy training with limited iterations)	✓ (Multiple reflection rounds)	✗ (Single correction phase)	✓ (Multi-round collaborative thinking)
Feedback Aggregation Mechanism	✓ (Centralized reflection pool with expert contributions)	✗ (No aggregation between iterations)	✗ (Majority voting only)	✗ (Requires answer labels)	✗ (No aggregation mechanism)	✗ (Manual collaboration strategies)
Applicability Across Domains	✓ (Modular architecture adaptable to various domains)	✗ (Medical-specific)	✓ (General but limited to reasoning only)	✓ (Limited to multiple-choice questions)	✗ (Math-specific)	✓ (General collaborative tasks)
Reasoning Transparency	✓ (Explicit reasoning trails with expert feedback)	✓ (Explicit reasoning loops)	✗ (Black-box improvement)	✓ (Explicit reasoning trails)	✓ (Explicit correction of reasoning)	✗ (Limited reasoning transparency)
Error Reduction Focus	✓ (Targeted reduction of hallucinations and inconsistencies)	✓ (Focus on medical hallucination reduction)	✗ (General performance improvement)	✓ (Reflection-based error reduction)	✓ (Mathematical error correction)	✗ (Focused on collaboration, not error reduction)
Scalability	✓ (Modular experts can be added/removed based on task)	✗ (Fixed methodology)	✗ (Limited to two iterations)	✗ (Fixed structure)	✗ (Domain-locked architecture)	✗ (Complex agent personality modeling limits scalability)
Real-time Adaptability	✓ (Dynamic feedback based on trial performance)	✗ (Predetermined loop structure)	✗ (Pre-trained model)	✗ (Requires answer labels)	✗ (Pre-trained error correction)	✓ (Adaptive collaborative strategies)

FIGURE 1. Comprehensive literature survey highlighting the key differences between our proposed architecture and related approaches for enhancing LLM reasoning capabilities.

Besides addressing challenges of hallucinations and insufficient contexts, there have been studies that investigate how well multiple LLM agents can work together. One such study investigates how different collaboration strategies and personality traits of LLM agents can be incorporated in a collaborative environment to bring about a targeted outcome [15]. The test-bed establishes agents with different personality traits (e.g., easy-going, confident) and employs debate and reflection thinking patterns in various collaborative strategies. However, it attempts to traverse all possible scenarios by search alone, which limits the agents' ability to adaptively make autonomous decisions on collaborative strategies in specific scenarios. Key findings from the evaluations over multi-round interactions of this test-bed suggest that specific combinations of similar and diverse thinking strategies seem to enhance the performance of a team of LLM agents that are required to perform on a common goal, highlighting human-like behaviors such as conformity and majority rule.

The proposed architecture differentiates itself from prior works by integrating a multi expert LLM feedback layer with multi-round reflective reasoning, this framework enables collaborative introspection by routing incorrect and halted queries through diverse expert LLMs. Compared to previous frameworks (refer to Figure 1), which either rely on CoT correction or domain-specific training (e.g., medical or math), the proposed system is more domain-agnostic and scalable across reasoning tasks. It also improves the architecture by allowing feedback without needing to manually define collab-

oration strategies. Instead, it builds a structured feedback loop using real-time trial evaluations, maximizing factual consistency and adaptability through reflections, while offering a principled method for refining LLM outputs over multiple passes.

III. PROPOSED APPROACH

This section outlines the training process architecture and methodology implemented for investigating our proposed solution.

Datasets:

We considered the following points to find the ideal datasets:

- **Multi-Step Reasoning:** The dataset should require LLM agents to perform tasks that cannot be solved directly. This challenges LLM agents to perform either convoluted or multi-step tasks in order to reach an answer.
- **Diverse Question Types:** In order to properly generalize our architecture, a dataset consisting of a broad range of questions should be used for evaluation. This provides the architecture with more versatile learning.
- **Contextual Information:** The dataset must incorporate a certain amount of contextual information for the agent to generate its own thoughts. This enables us to test the amount of relevant information an agent requires in real scenarios to generate an appropriate response.
- **Real World Relevance:** The dataset should contain

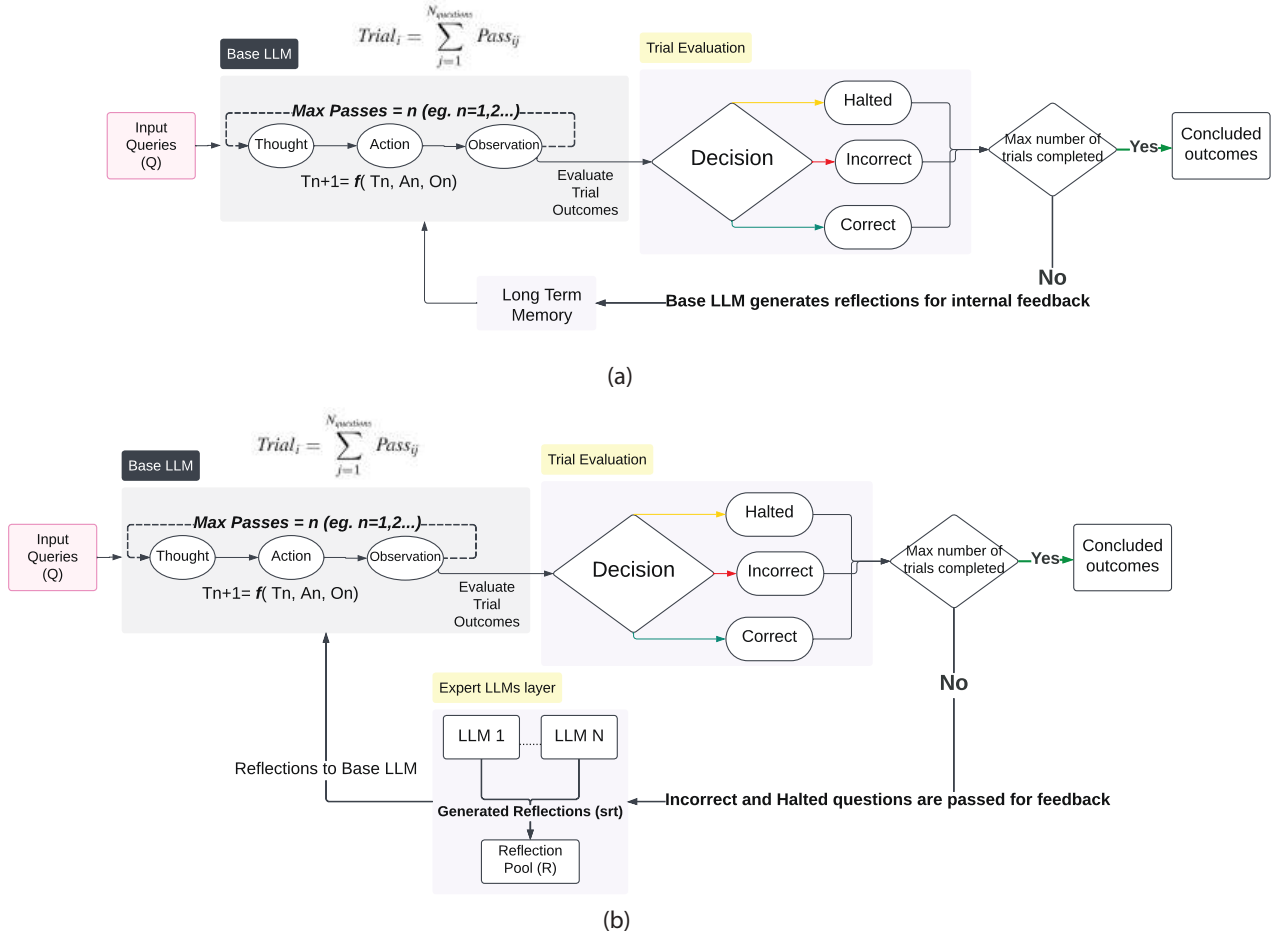


FIGURE 2. (a) Baseline architecture combining Reflexion + ReAct agent showing the flow of self-reflection with internal feedback; (b) Proposed architecture, where incorrect and halted outcomes trigger the layer of expert LLMs to generate a set of reflections for the reflection pool utilized by Base LLM for refinement.

natural-language queries that mimic real-world scenarios, making our architecture more suitable for inference time compared to current approaches.

HotpotQA, SimpleQA, and PubMedQA datasets have been used for the training and testing method. Detailed descriptions for these are provided below:

- **HotPotQA dataset [5]:** It is a question-answering dataset containing natural language questions that require multi-step reasoning to answer. The dataset includes multiple question-answer pairs, each accompanied by two short extracts of supporting facts from Wikipedia. The construction of this dataset has been done by an NLP research team at Carnegie Mellon University, Stanford University, and Université de Montréal. Tailored with strong supervision, the dataset only provides specific hints with question-answer pairs to ensure that the LLM agent does not fully rely on these supporting facts to generate factually correct answers. Regarding this study, the multi-hop reasoning challenges in the dataset allow us to measure how effectively our

architecture can infer and synthesize information. Additionally, the diverse question types in this dataset help to evaluate the architecture's ability to handle question styles with different reasoning pathways.

- **SimpleQA Dataset [16]:** This is a question-answering dataset, created by the team at OpenAI, that focuses on short and fact-seeking queries. The main idea behind this dataset is to make measuring factuality more controllable. The answers to questions in this dataset are singular and indisputable, making it easier to grade. It maintains diversity by including synthetic questions modeled from real-world queries that range from science and technology to TV shows and video games. To ensure consistent evaluation with minimal variance, this corpus only contains 4,326 questions with high correctness in reference answers that is supported by two independent AI trainers. This allows us to measure the impact of calibration on performance measurement when combined with appropriate evaluation metrics. Furthermore, the short queries and definitive answers in SimpleQA emphasize factual brevity that helps distinguish when

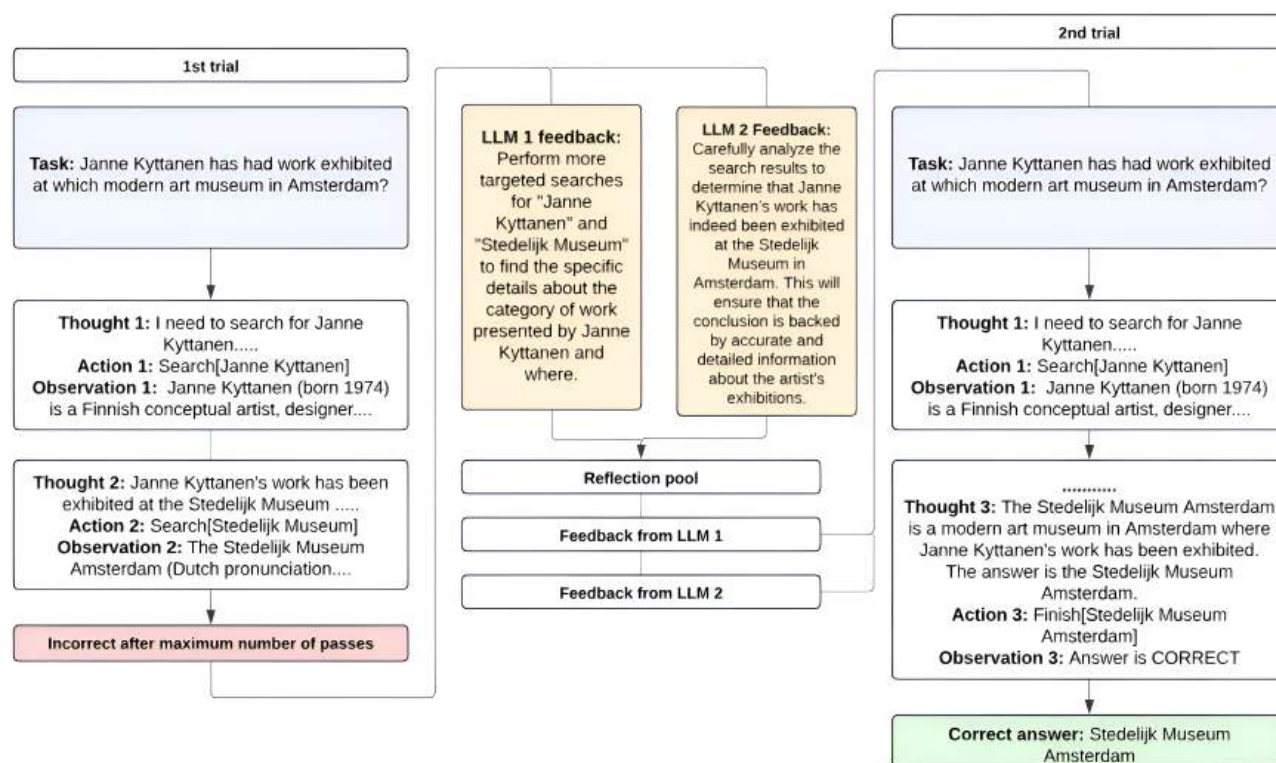


FIGURE 3. Flow of reasoning process of the architecture incorporating Reflexion + ReAct agent on HotPotQA dataset. The core flow of generating feedback and integrating reflection pools remains common while dealing with each dataset.

the architecture produces a guess or an incomplete response. This aligns with effectively evaluating how the proposed architecture can handle direct queries and refine it through iterations.

- **PubMedQA Dataset [17]:** This is a specialized dataset that has been designed for question-answering tasks in biomedical research. This dataset contains 1K expert-labeled, 61K unlabeled, and 211K artificially generated question-answer pairs. Each of the questions must be answered with 'yes', 'no', or 'maybe', using the passed context. Each instance of PubMedQA consists of a question from an existing research or from articles drawn from PubMed Central [18] or PubMed abstracts [19], an appropriate abstract without a direct answer, a long answer that contains the final answer and conclusion with respect to the abstract, and a yes/no/maybe label that is used to summarize the conclusion. Some of the clinical study-related topics used in the reference articles for the questions are: "Pregnancy Outcome," "Survival Analysis," and "Case-Control Studies." The dataset's ability to support evidence-based clinical decisions makes it ideal for testing how closely our architecture can generate the required conclusions. PubMedQA deals with biomedical literature that requires reasoning about clinical or research-based topics, making it suitable to assess the domain adaptability of the architecture along with how effectively it handles evidence-based inference and

clinical concepts. The response format of this dataset expects a 'yes,' 'no,' or 'maybe' conclusion helping to assess the model's ability to differentiate between certainty and ambiguity. Moreover, the response summary requires robust factual consistency that tests the model's ability to deal with specialized scientific texts and handle technical terms.

Note: While these are widely adopted benchmarks in multi-step and reasoning tasks, several studies demonstrate that biases and limitations in understanding natural language tasks still persist. For instance, research indicates that questions based on adversarially modified paragraphs has been shown to severely degrade the performance of several models [20]. To address such vulnerabilities, we outline some future strategies in the Future Work and Conclusion section.

IV. MODEL ARCHITECTURE

A. ARCHITECTURE WITH REACT AGENT

The proposed architecture adopts a multi-tiered approach to enhance LLM capabilities by layering a variety of targeted feedback.

At its core, a pre-trained LLM, such as GPT-3 [21] or its successors, serves as the foundational model for generating text. Treating each question as a separate environment (since every question requires a different context), the base LLM undergoes a thought generation step (defined as a single pass on that question) of generating a reasoning trail over each

Expert LLM	Primary Function	Selection Criteria	Domain Expertise	Computational Profile	Complementary Strengths
Claude 3-haiku	Deep contextual reasoning and detailed inference	Designed for handling complex, context-heavy problems requiring multi-step reasoning	General domain with strong performance on reasoning tasks requiring conceptual understanding	Higher computational requirements with superior generative depth	Human-like reasoning capabilities with ability to capture conversational patterns and produce elaborate responses
Minstral 8B	Factual precision and rapid inference	Compact architecture with precision during rapid inference	General domain with strength in factual reasoning and short, targeted queries	Lightweight model allowing for easier scalability and deployment	Efficient processing of short, fact-seeking queries with high precision and minimal hallucination
MedAlpaca	Medical domain expertise	Fine-tuned specifically for medical reasoning tasks	Specialized in medical terminology, diagnoses, and clinical reasoning	Moderate computational requirements with domain-specific optimization	Highly accurate responses aligned with clinical guidelines; specialized medical knowledge comparable to medical students
PMC-LLaMA	Biomedical research comprehension	Refined using PubMed Central for scientific literature analysis	Expertise in diagnostic tasks and complex medical pathways	Moderate to high computational needs balanced with domain efficiency	Strong capability in analyzing research findings and correlating experimental data within medical contexts

FIGURE 4. Functional roles and selection criteria for each expert LLM used in our study. The expert selection was based on evaluation across task suitability, model efficiency, reasoning depth, and domain adaptability. In the current study, there is no inter-model dependency or collaboration between experts, with each expert functioning independently and feedback aggregation occurring at the reflection pool level.

environment.

$$Thought(T_n) \rightarrow Action(A_n) \rightarrow Observation(O_n)$$

$$T_{n+1} = f(T_n, A_n, O_n) \text{ (Base LLM)} - (1)$$

where function f is responsible for updating thought (T_{n+1}) based on the previous Pass,

$$Pass_{ij} = T_j \rightarrow A_j \rightarrow O_j - (2)$$

This, when constituted with the maximum number of passes (in this case, a maximum of 6 passes) per environment, accounts for a single trial.

$$Trial_i = \sum_{j=1}^{N_{questions}} Pass_{ij} - (3)$$

In the three-step process, each thought is generated by extracting the core information either from the original query within datasets or from the observations generated by the LLM. Having the core information at disposal, the LLM selects an action type among Search (to retrieve general information), Lookup (to retrieve specific information) or Finish (if reached a definite answer) to use. If the action type is among Search or Lookup, then the retrieved information is called an observation, which consists of a summary of information drawn from Wikipedia articles with the help of its API provided to the ReAct agent.

In one trial, the LLM can come to only one of the three conclusions C for an environment. These include Correct

(answers that exactly match the target labels; see Appendix), Incorrect (answers that do not exactly match the target labels; see Appendix), and Halted (where the model is unable to reach a definite/conclusive answer or requires more passes for the question). These conclusions are drawn if the agent returns the 'Finish' action choice for a question or after completing the maximum number of passes in a single trial for each question.

$$C_i \in \{Correct, Incorrect, Halted\} - (4)$$

where C is concluding outcome of the trial.

After completing a single trial (maximum number of passes reached for every question), all details about the incorrectly answered and halted questions go to a layer of "expert-in-their-field" LLM agents. The integration of this expert LLM layer ensures that the base LLM extends beyond its general knowledge and draws feedback from specialised LLMs as required (Note: For a visual overview of the process of integrating LLMs in the architecture, refer to Figure 2). The expert LLM layer then generates reflections, summarizing the key points and lessons learned in the form of a reflection pool R_i . This reflection pool consists of all the feedback generated by the expert layer for the current trial with respect to each reasoning trail of the base LLM.

To specialize in specific domains, a collection of expert LLMs is integrated, providing domain-specific knowledge and relevant insights (eg. MedAlpaca [22] for medical, MetaMath [23] for mathematics and AlphaCode [24] for programming). These experts are only provided with details of the

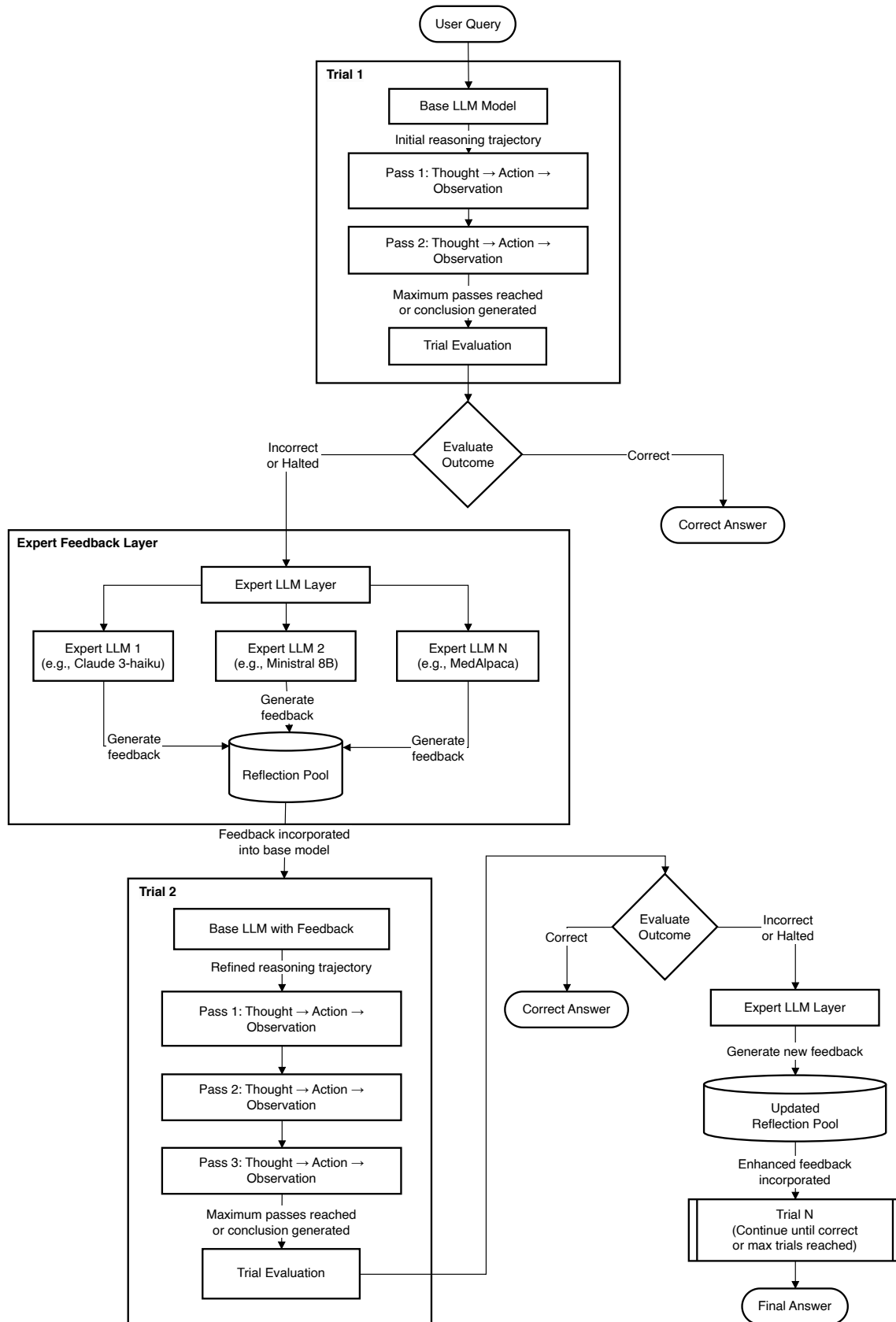


FIGURE 5. Detailed workflow of the iterative refinement process in the proposed brokered multi-expert reflection approach. The flowchart illustrates the complete cycle from initial user query through multiple trial iterations.

trial (reasoning trails for questions) if there are any subsequent trials left, after which the expert layer engages with the reasoning trails to generate feedback.

After the conclusion of a single trial and feedback generation step, the base LLM is required to generate new reasoning trails based on feedback in the reflection pool. This carries on until the maximum number of trials is reached, after which the training process is concluded.

$$LLM_{i+1} = LLM_i + g(R_{i+1}) - (5)$$

where g is the function for sending feedback from the reflection pool for the relevant task.

By iteratively refining the LLM's responses and incorporating expert knowledge on incorrect and halted questions, it structures the whole process to resemble reinforcement learning with feedback-driven policy refinement, resulting in a robust and versatile LLM agent.

As seen in Figure 2(a), the baseline architecture combining the *Reflexion + ReAct* agent may reinforce errors due to its sole reliance on self-generated reflections. In contrast, the proposed method replaces this internal feedback mechanism for an expert layer that generates multiple varieties of feedback over the reasoning trails of the base agent. Additionally, the integration of the expert layer allows the architecture to have modularity while selecting the expert LLMs based on requirements and adapt much more efficiently without retraining the entire model. The modularity and centralization of external feedback for expert LLMs allows the proposed architecture in Figure 2(b) to be more flexible and robust.

Figure 5 illustrates the iterative refinement process in the proposed brokered multi-expert reflection approach. The process begins with the base LLM generating an initial reasoning trajectory (Trial 1), which undergoes evaluation. If the outcome is incorrect or halted, the expert feedback layer engages multiple expert LLMs to analyze the reasoning path and generate targeted feedback. This feedback is aggregated in a reflection pool and incorporated back into the base LLM for subsequent trials. The system continues this refinement cycle, progressively improving reasoning quality until either reaching a correct answer or exhausting the maximum allowed trials.

Building upon the *Reflexion + ReAct* agent, our pseudo-algorithm to combine the feedback of multiple experts is as follows:

Initialize all LLMs and reflection pool.

For each $Trial_i$:

- Expert LLMs generate feedback sr_t for the reasoning trails of *Incorrect* or *Halted* questions from previous trials
 - The updated reflection pool R_{i+1} is created by $R_{i+1} = sr_t$ if feedback generation step is done previously
 - For each question in the dataset, perform the maximum number of passes ($Pass_{ij} = T_j \rightarrow A_j \rightarrow O_j$)
 - Trial evaluation step $C_i \in \{Correct, Incorrect, Halted\}$
- Base LLM is provided with relevant feedback of a task from

Reflection pool R_{i+1} using $g(R_{i+1})$.

Repeat the process until the maximum number of trials (n) is reached.

B. MODIFIED ARCHITECTURE WITH COT AGENT

For testing the reasoning only abilities of the architecture in domain-specific tasks of the PubMedQA dataset, we adopt *Reflexion + Chain-of-Thought (CoT)* with context. The difference between the main architecture that uses *Reflexion + ReAct* agent with respect to the *Reflexion + Chain-of-Thought (CoT)* is only between how the pass is conducted across each environment. Since PubMedQA questions already contain the sources from which they are drawing the context, utilizing the CoT agent seems a viable option where all the reasoning context is passed as ground truth C_{gt} instead of giving the agent access to the Wikipedia API. For each environment $Q, C_{gt} \rightarrow A$, process is implemented, where Q is the task and C_{gt} is the context for solving the question, the pass is conducted in the following manner:

$$Pass_{ij} = T_j \rightarrow A_j \rightarrow O_j$$

where *Finish* is the only action choice provided to the agent and the observation step is used to determine whether the answer concluded is correct or incorrect. In this architecture, we only need one pass per question in each trial since the action choices and observations do not affect the thoughts generated by the *Reflexion + CoT* agent.

This slight tweaking of the base Agent makes it appropriate to deal with the reasoning-only environment of the PubMedQA dataset.

V. EXPERIMENTS

To evaluate the effectiveness of our proposed architecture, we conducted a series of experiments on a diverse range of question-answering benchmarks. As a baseline, we adopted the solutions proposed by Shinn, Noah, *et al.* [25] that use *Reflexion + ReAct* [26] and *Reflexion + Chain-of-Thought (CoT)* [27]. To evaluate the performance of the proposed architecture on questions that require reasoning and action choice, we implemented *Reflexion + ReAct* agent that has access to Wikipedia API to retrieve relevant contextual information in the HotPotQA and SimpleQA datasets. Additionally, *Reflexion + Chain-of-Thought* is implemented to test reasoning-only capabilities on domain-specific tasks of the PubMedQA dataset. These agents perform strongly in reasoning and action choice tasks. GPT-3 served as a foundational pre-trained LLM to generate text in the baseline method. This baseline model was enhanced by incorporating our proposed architecture, enabling it to leverage an expert knowledge layer in a cyclic reinforcement learning manner to further improve its reasoning capabilities.

Figure 3 illustrates the flow of reasoning process of the architecture, which incorporates *Reflexion + ReAct* agent in the HotPotQA dataset. The core flow of generating feedback

Model	ROUGE-L	Consistency(CTRL-Eval)
HotPotQA		
Baseline with Ministral 8B feedback	0.543	-5.245
Baseline with Claude 3-haiku feedback	0.494	-5.204
Ours with Ministral 8B + Claude 3-haiku feedback	0.577	-5.114
SimpleQA		
Baseline with Ministral 8B feedback	0.220	-5.118
Baseline with Claude 3-haiku feedback	0.214	-4.903
Ours with Ministral 8B + Claude 3-haiku feedback	0.217	-4.689

FIGURE 6. Quantitative evaluation of our approach with different variations for HotPotQA and SimpleQA Dataset using the evaluation metrics ROUGE-L and Consistency Score (CTRL-Eval)

Model	Correct	Incorrect	Halted
HotPotQA			
Baseline with Ministral 8B feedback	46	21	33
Baseline with Claude 3-haiku feedback	45	6	49
Ours with Ministral 8B + Claude 3-haiku feedback	48	12	40
SimpleQA			
Baseline with Ministral 8B feedback	18	18	64
Baseline with Claude 3-haiku feedback	14	17	69
Ours with Ministral 8B + Claude 3-haiku feedback	14	8	78

FIGURE 7. Overview of the number of answers that were concluded incorrect, correct and halted after 5 subsequent trials during the trial evaluation phase. (Halted: If the model was not able to reach a definite answer within a given maximum number of passes in a trial then it was passed in the subsequent trials for a re-attempt (assuming there are trials left)).

and integrating reflection pools remains consistent across datasets.

For each dataset, including HotPotQA, SimpleQA, and PubMedQA, we limit our experimentation to 100-pair subsets to conserve computational resources and allow multiple iterative trials. While the findings we present provide an in-depth analysis of the expert layer, they may be less representative of the full datasets, and we acknowledge that scalability and generalizability to larger samples remain areas for further investigation.

Note: This study does not include comparative study no vs single expert feedback (i.e., evaluating the base agent alone), as prior work by Renze et al. [13] has already established that incorporating even a single expert—via self-reflection or external guidance—consistently improves LLM performance over models without any expert feedback. Therefore, the comparative analysis in this study focuses on evaluating the per-

formance of the proposed multi-expert architecture against configurations involving single expert feedback layers, to assess the impact of combining complementary reasoning perspectives.

Expert LLMs have been selected to provide complementary strengths for reasoning across general and domain-specific tasks. Their functional roles and selection criteria are summarized in Figure 4. To examine our approach, we divide our experimentation into two sections:

A. GENERALIZED REASONING TASKS

This Section elaborates on our approach while dealing with datasets that have generalized reasoning tasks in order to evaluate the model's capability to deal with limited context, reduce hallucinations, and develop diverse semantic understanding of environments. We used HotpotQA and SimpleQA Dataset for the evaluation process.

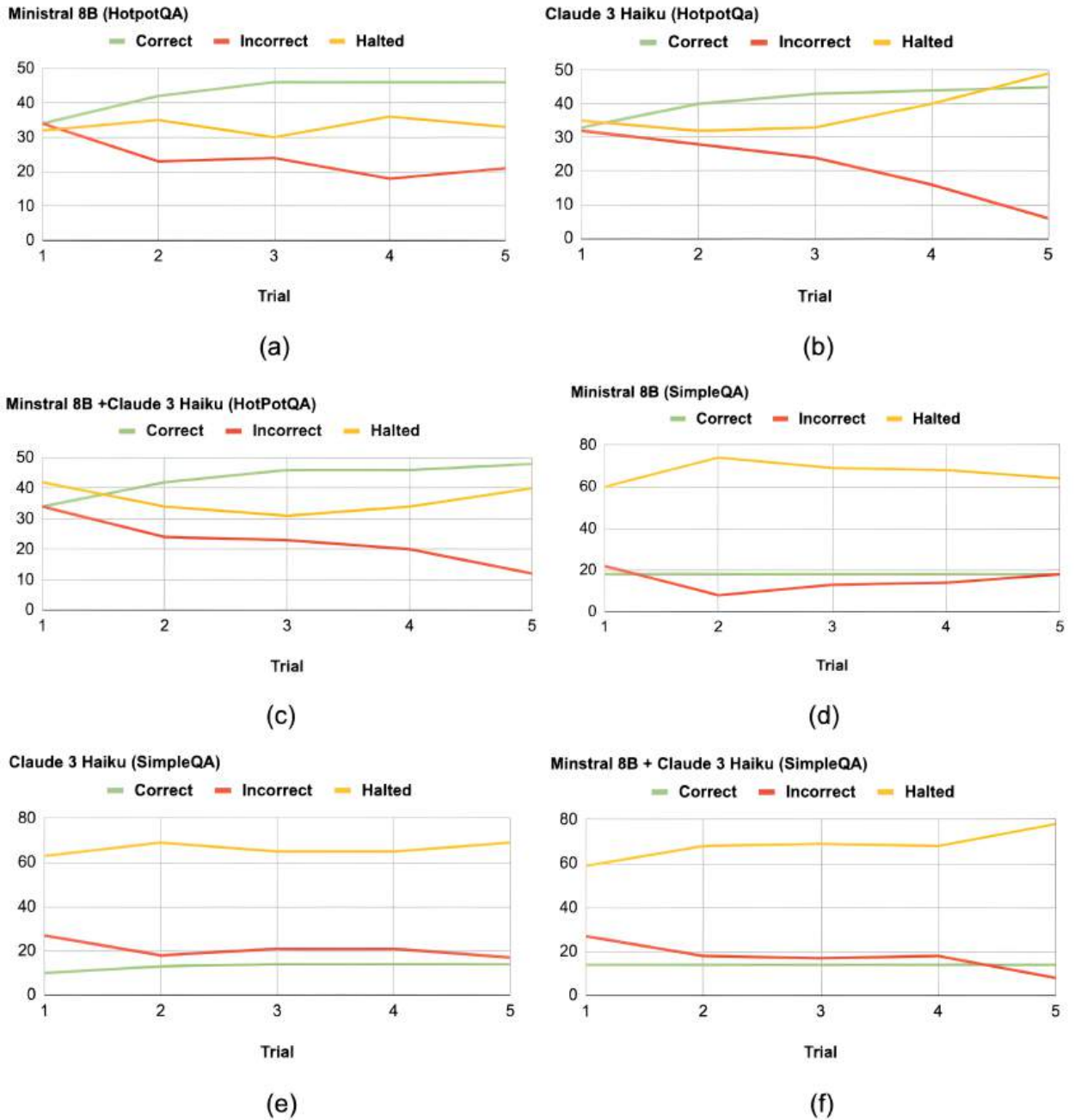


FIGURE 8. Success signals of the proposed architecture with single and combined expert feedback over 5 trials on HotPotQA and SimpleQA datasets

A combination of expert LLMs that makes up a robust feedback layer while handling a wide variety of reasoning tasks from both the datasets has to be both efficient and profound. To address this, we carefully select each expert based on its unique capabilities and functional role within the architecture. We take advantage of the lightweight model size of Minstral 8B [28], which allows us to take advantage of its efficiency and scalability in short-form, factual reasoning tasks such as those in SimpleQA. The rationale for the selection for Minstral 8B is detailed in Figure 4. To address

the broader, complex, and context-heavy nature of multi-step tasks in HotPotQA, we also incorporate Claude 3-haiku [29] that is inherently designed to deal with long and complex contexts. (see Figure 4)

B. DOMAIN SPECIFIC REASONING TASKS

In order to examine how well our proposed solution handles domain-specific tasks, we employ our architecture on the PubMedQA Dataset [17] dataset, which we use as a benchmark, particularly to test how the architecture generates

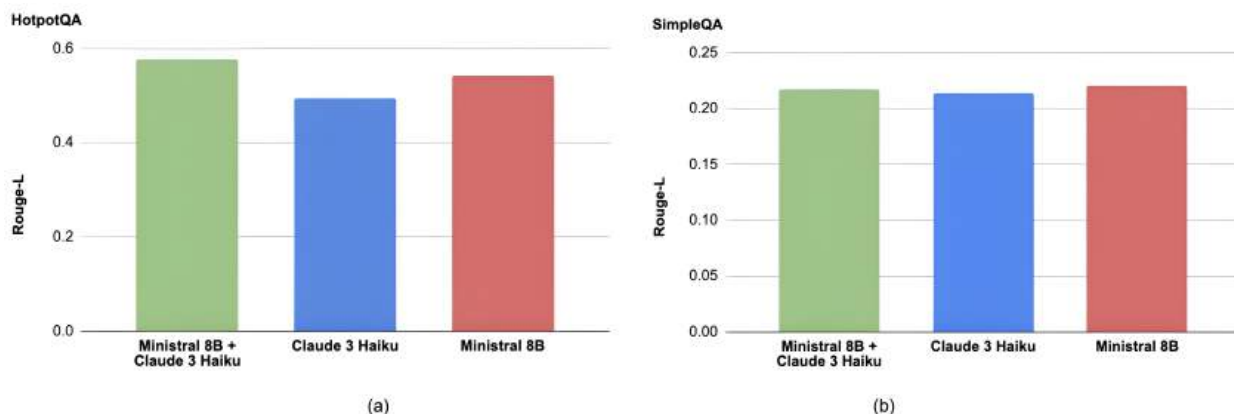


FIGURE 9. Rouge-L scores of the proposed architecture with single and combined expert feedback after completing 5 trials on HotPotQA and SimpleQA datasets

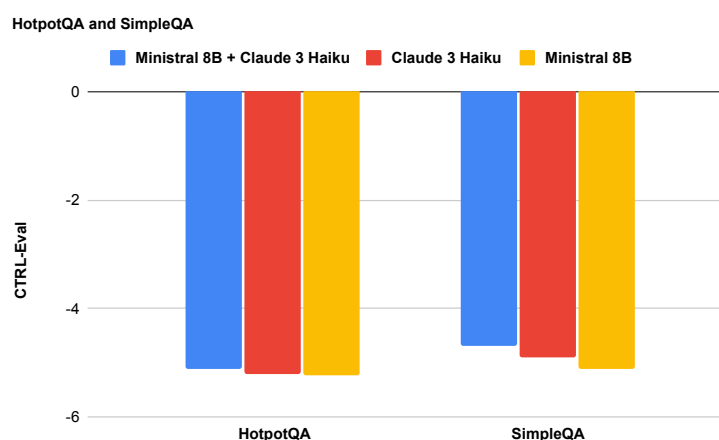


FIGURE 10. Consistency (from CTRL-Eval) scores of the proposed architecture with single and combined expert feedback after completing 5 trials on HotPotQA and SimpleQA datasets

semantically appropriate conclusions in the bio-medical research field. To incorporate medical LLM experts, we utilize fine-tuned medical reasoning models MedAlpaca [22] and PMC-LLaMA [30]. These models are integrated in our expert layer to give feedback on the actions carried out by our base Reflexion + CoT agent. To handle the comprehensive PubMedQA dataset effectively, we selected a subset of question-answer pairs encompassing a diverse array of medical topics, including prospective studies, risk assessment, and case-control studies, among others.

The principal function of MedAlpaca and PMC-LLaMA is to provide specialized feedback with the aim of ameliorating false positives and false negatives generated by the base LLM. This function is highly significant within the medical domain, given its inherent emphasis on precision and efficiency. The utilization of MedAlpaca in this framework is advantageous for its strong alignment with clinical guidelines and specialized medical training (refer to Figure 4). However, acknowledging the limitation of MedAlpaca to the medical domain and its knowledge being commensurate with that of

medical students, it became imperative to integrate another expert LLM, PMC-LLaMA, to broaden the focus spectrum. PMC-LLaMA is an LLM, refined using PubMed-Central, endowed with the capability to tackle diagnostic tasks and comprehend intricate medical pathways. For inferential purposes, PMC-LLaMA is adept for analyzing research findings and correlating experimental data. Concerning integration within our architecture, PMC-LLaMA rigorously assesses the reasoning provided by the base model in conjunction with the observations made by the model, ensuring the relevance of attributes to the medical domain. This diverse focus offered by both expert models assists in addressing elements such as coherence and semantic pertinence in our model when addressing medical reasoning tasks.

Note: The medical models MedAlpaca and PMC-LLaMA have not been tested in real-world applications, so they should never be used as a substitute for a doctor's opinion and must be treated as a research tool only.

VI. EVALUATION METRICS

To evaluate the multi-tier architecture of our approach, we employ a diverse set of metrics including ROUGE-L [31] and CTRL-Eval [32]. ROUGE-L serves as an evaluative measure for determining the quality and precision of responses produced by the language model, by analyzing the congruence between actual ground-truth answers and the generated outputs. Additionally, we employ CTRL-Eval as a metric to assess the efficacy of controlled text generation. This thorough compilation of metrics facilitates a comprehensive assessment of the language model's performance, offering an in-depth depiction of its output in terms of quality, relevance, and structural coherence.

Utilizing these evaluation metrics, this study seeks to elucidate the capabilities of our architectural framework in addressing intricate multi-step reasoning challenges. A detailed exposition of these evaluation metrics follows in the subsequent sections.

A. ROUGE-L

To assess the quality of a summary comparative analysis is conducted between the generated summary and other reference summaries crafted by human authors. This involves examining the overlap of n-grams, word sequences, and word pairs. Among the measures employed, ROUGE-L is particularly notable for its emphasis on the longest common subsequence between two textual sequences. The utilization of ROUGE-L (Sentence-Level LCS) in our research offers significant advantages as it prioritizes semantic similarity between sentences rather than merely depending on word-matching mechanisms. Moreover, ROUGE-L requires relatively straightforward computation and is capable of accommodating various sentence structures. Consequently, it emerges as an optimal metric for evaluating the performance of our architectures in text-generation reasoning tasks.

$$F_{lcs} = \frac{(1 + \beta^2)R_{lcs}P_{lcs}}{R_{lcs} + \beta^2P_{lcs}}$$

This equation is used to calculate the summary-level ROUGE-L F-measure using R_{lcs} (recall at summary level), P_{lcs} (precision at summary level) and β (this parameter controls the weighting between recall and precision).

B. CTRL-EVAL

The evaluation approach employed integrates text-infilling tasks and probability fusion to derive final scores concerning specific aspects such as coherence, consistency, and attribute relevance. In this study, we utilized the consistency dimension of this function to determine if the model can produce answers that are consistent with the content prefix. Given that this metric assesses the consistency between the content prefix and the generated text, it serves as a suitable evaluation function to investigate a wide spectrum of text-generation scenarios in our experiments. Furthermore, this metric exhibits robust generalization capabilities when applied to models of varying qualities, thereby providing insights into potential model

performance in realistic scenarios as opposed to controlled environments.

$$S_{cons}(I) = NISF(Y_{\setminus x}).\log P_{\theta}(Y_{\setminus x}|X[M]) + NISF(X).\log P_{\theta}(X|[M]Y_{\setminus x})$$

CTRL Eval Consistency score is calculated by taking the *log probability* of a pre-trained model as the evaluator's score and then weighting them with normalized inverse sentence frequency where $Y_{\setminus x}$ denotes the remaining part of the generated text without the prefix.

Although this study relies mainly on N-gram-based metrics (e.g., ROUGE-L) and consistency scores, we recognize that BERTScore, MoverScore, and BLEU can offer additional insights into semantic similarity and precision. However, due to computational constraints while focusing on consistency and factual alignment, the current methodology does not integrate these metrics. As discussed in the Future Work Section, we plan to explore these complementary metrics to provide additional insights of summary quality to future readers.

VII. RESULTS AND COMPARISONS

The performance of the baseline model and our proposed architecture has been compared on tasks necessitating Complex Reasoning (involving questions that require multiple steps of reasoning and inference) and Domain-Specific Knowledge, particularly within the realm of biomedical research. The evaluation of model performance on these task categories has been undertaken to ascertain the influence of our proposed architecture on the large language model's capacity to produce responses that are more accurate, informative, and contextually pertinent. Given the inherent subjectivity present in natural language responses, our evaluation of trial results for reasoning tasks was conducted using exact match metrics for the agent's responses. To further account for linguistic variability while ensuring the provision of conclusive, rather than purely verbatim answers, our analysis also integrated Rouge-L and Consistency scores. These metrics assess the models' proficiency in extracting essential information and delivering semantically relevant abstracts.

A. EVALUATION ON HOTPOTQA

1) Analysis

Figure 6 suggests that the proposed architecture generates better Rouge-L (0.577) and Consistency (-5.114) scores when using combined feedback from Claude 3-haiku and Ministral 8B compared to the baseline using individual feedback. Our proposed architecture outperforms all the baselines for the HotPotQA dataset, and this can be attributed to the complementary strengths of the expert models which provided a comprehensive and subtle reasoning guide for the base model. A higher Rouge-L score (Figure 9 (a)) suggests that the responses generated by the model align with reference answers closely while maintaining high fluency and relevance. This leads to an improvement in overall recall, requiring key components of the reference text to be included in the response. Additionally, a lower (closer to 0) Consistency Score

PubMedQA		
Model	ROUGE-L	Consistency(CTRL-Eval)
Baseline with PMC-LLaMA feedback	0.238	-1.804
Baseline with MedAlpaca feedback	0.220	-1.840
Our approach with PMC-LLaMA + MedAlpaca feedback	0.244	-1.719
Our approach with Ministral 8B + Claude 3-haiku feedback	0.229	-2.110

FIGURE 11. Quantitative evaluation of our approach with different variations for the PubMedQA Dataset using the evaluation metrics ROUGE-L and Consistency Score (CTRL-Eval)

PubMedQA			
Model	Correct	Incorrect	Inconclusive
Baseline with PMC-LLaMA feedback	61	12	27
Baseline with MedAlpaca feedback	61	12	27
Our approach with PMC-LLaMA + MedAlpaca feedback	59	9	32
Our approach with Ministral 8B + Claude 3-haiku feedback	69	15	16

FIGURE 12. Overview of the number of answers that were concluded yes, no or maybe accordingly on the PubMedQA tasks. (Correct: if the answer correctly consist of a Yes, No or Maybe as the final decision label before the concluding statement, Incorrect: if the answer incorrectly mentions the decision label before the concluding statement, Inconclusive: if the pass is halted or doesn't contain a final decision label)

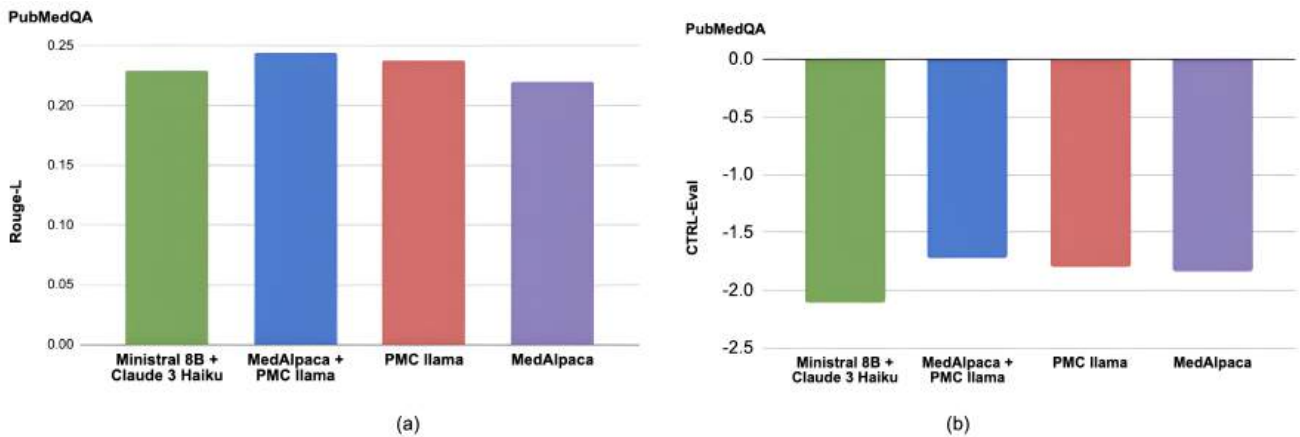


FIGURE 13. (a) Visualization of the Rouge-L metric scored by all the variations of the architecture on the PubMedQA dataset. (b) Visualization of the Consistency metric from CTRL-Eval scored by all the variations of the architecture on the PubMedQA dataset

(Figure 10), which comes from CTRL-Eval, indicates the model's ability to generate more factually accurate responses that are semantically consistent with the content prefix.

Figure 8 (a), (b) and (c) and Figure 7 depict an improved overall accuracy of the model in dealing with HotPotQA reasoning tasks over 5 subsequent trials. It highlights the trend that as the number of trials increases, the number of responses generating a correct success signal will increase and the number of responses generating an incorrect success signal will reduce. This trend is visually represented in Figure 8 (a), (b) and (c) when the base model is integrated with the

iterative feedback mechanism of our architecture. When using combined feedback, the proposed architecture showcases the superior ability to leverage expert feedback. This difference in performance can be credited to the wider set of errors that the multi-expert layer is able to identify.

Halted questions represent the unanswered questions (after maximum passes in a single trial) that were re-attempted in the next trials. Figure 8 (a) observes a fluctuating pattern in halted questions that indicates the potential struggle of Ministral 8B while dealing with questions that require more intricate reasoning where individual Ministral 8B feedback

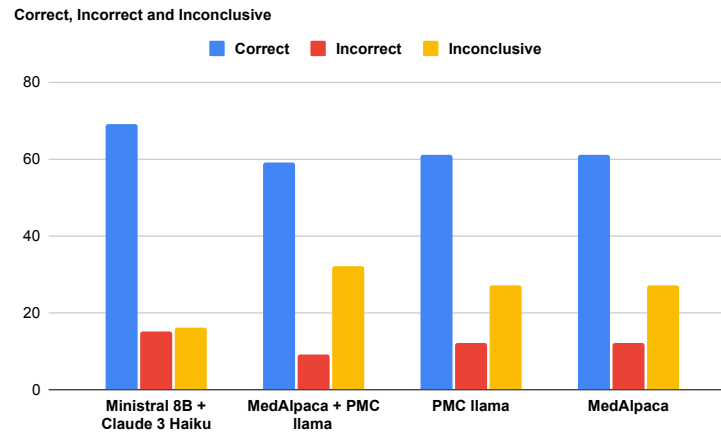


FIGURE 14. Visualization of the number of answers that were concluded yes, no or maybe accordingly on the PubMedQA tasks

seems to be less effective. Along with this, a slight decrease and eventual increase in the number of halted questions in Figure 8 (b) for Claude 3-haiku suggests an adjustment in the reasoning approach of the Claude 3-haiku model. Being a deeper reasoning model, Claude 3-haiku might showcase achieving quick conclusive feedback in the initial trials (hence, the slight dip); however, while dealing with complex tasks, it can potentially explore different reasoning trajectories before reaching any concrete feedback (hence, the increase). In Figure 8 (c), the combined feedback layer leads to an initial decrease of the halted questions that eventually stabilizes around the initial number after an increase. The initial dip and subsequent increase represent the rapid adaptation and active exploration of different feedback trajectories. Eventually, the number of halted questions started to stabilize around 40, which indicates that the expert layer had achieved a balanced combination of expertise that leverages complementary strengths. This is a positive indicator for the architecture's ability to adapt and generate robust feedback throughout an iterative process.

Moreover, the proposed method utilizing feedback from Ministral 8B+ Claude 3-haiku excels by demonstrating a consistent upward trend (Figure 8 (c)) in generating the highest number of Correct responses over multiple subsequent trials, as seen in Figure 7. The number of incorrect responses also steadily decreases for the same model as compared to the trends observed for individual feedback variations in Figure 8 (a) and (b). This observation suggests that as the expert layer gets familiar with the dataset, the feedback improves, resulting in lower incorrect answers and higher correct answers.

2) Results

The integration of the expert feedback layer within the proposed architecture led to an improvement in performance in terms of higher success signal, along with an improvement in Rouge-L and Consistency Scores, as evidenced by Figure 7 and Figure 8 (c). This highlights the effectiveness of our

approach on HotPotQA dataset while dealing with multi-step reasoning environments. These results support the argument that by adding an expert layer of LLMs on top of a reasoning model, we can effectively increase the performance of the model for generating factually consistent responses in multi-step reasoning tasks.

B. EVALUATION ON SIMPLEQA

1) Analysis

Figure 6 and Figure 10 demonstrate that the Consistency Score (-4.689) associated with the SimpleQA dataset shows a significant improvement in our proposed architecture compared to the baselines (-5.118 and -4.903). This enhancement indicates that the model successfully upholds a consistent methodology when addressing questions subject to diverse interpretations. Furthermore, the model's ability to distill fundamental meaning while preserving semantic integrity to produce reliable responses aligns with the intended purpose of the SimpleQA dataset.

Rouge-L scores across all the models are relatively similar in Figure 6. Achieving a Rouge-L score of 0.217 with our proposed method, when compared to the baseline scores of 0.220 and 0.214, underscores its competitive potential. This approach not only sustains a comparable Rouge-L score but also enhances the Consistency Score, even on the more challenging question sets from the SimpleQA dataset. Such performance is particularly beneficial for the SimpleQA dataset, which prioritizes factual accuracy and brevity in generated responses. This equilibrium between Rouge-L and Consistency Scores indicates that the proposed architecture effectively integrates expert feedback, resulting in discernible improvements by comprehending semantic alignment and generating responses that are factually accurate.

Figure 7 and Figure 8 (d), (e) and (f) provide a breakdown of how the different variations of the architecture go over five subsequent trials and lead to different numbers of correct, incorrect and halted questions. Figure 7 indicates that although

the number of correct responses for the proposed method is 14, which is lower than that of the Ministral 8B approach (18 correct responses), the number of incorrect responses remains the least (8 incorrect responses) with respect to each variation, which is evident in Figure 7 along with the downward trend in Figure 8 (f). This highlights a key potential of our approach where the model might not be able to generate a correct answer but it reduces the number of incorrect answers generated, which is particularly valuable for the intended purpose of the SimpleQA dataset that emphasizes accuracy and conciseness.

Figure 8 (d) and (e) establish an important baseline which helps to understand the adaptation of individual models on questions from the SimpleQA dataset. In Figure 8 (d), the fluctuation in incorrect responses depicts an inconsistent trend, suggesting that the standalone model struggles with self-reflection over multiple iterations. While Figure 8 (e) illustrates that Claude 3 Haiku follows a similar trend to the combined feedback approach Figure 8 (f) on the SimpleQA dataset, the higher number of incorrect questions for Claude 3 Haiku highlights the weakness of the individual model in controlling error propagation.

This visualization suggests a trend for our architecture for halting certain questions rather than leading to a potentially incorrect response. Having the highest number of halted questions (see Figure 7) rather than incorrect ones provides insights into model's need for different or longer reasoning trails when dealing with the novelty of the SimpleQA dataset to which the LLMs might have limited exposure in terms of specific characteristics. Moreover, as the expert layer gains more experience with an increased number of trials (more than 5 trials) on the SimpleQA dataset, the feedback from them can become more specifically tailored, potentially leading to lesser number of halted questions and higher correctness [33]. This encourages the system to halt when there is insufficient information available and thereby reduce the number of possible incorrect conclusions for questions with unfamiliar contexts.

2) Results

The analysis of our proposed experiment on the SimpleQA dataset reveals the potential of the multi-tiered architecture even in handling a fairly intricate and challenging dataset. Maintaining a competitive score in both evaluations (Rouge-L: 0.217 and Consistency: -4.689) attributes to the model's focus on generating semantically appropriate responses. The observed values (Correct: 14, Incorrect: 8, Halted: 78) in Figure 7 reveal the model's objective of a more cautious approach and generating logically sound outputs which have further potential to be enhanced for applications that require reliability and precision. The straightforward nature of SimpleQA's questions often provide limited context, creating scenarios where the model struggles to retrieve or infer enough supporting information. It has been observed that when the architecture deals with a lack of alignment in context then it is more likely to stop. This dynamic is especially critical

to account for the short-answer contexts of the SimpleQA dataset that limits the inference ability of the architecture.

C. EVALUATION ON DOMAIN SPECIFIC REASONING TASKS

1) Analysis

Figure 11 demonstrates that the proposed architecture achieves higher Rouge-L score (0.244) and better Consistency score (-1.719) using combined feedback from PMC-LLaMA and MedAlpaca compared to baseline variations of the architecture that used individual expert feedback. To further test the robustness of the architecture, we also integrate the generalized reasoning LLM experts (Ministral 8B + Claude 3-haiku) for testing with PubMedQA tasks and Figure 11 shows that this variation also provide competitive scores (Rouge-L: 0.229 and Consistency: -2.110), however, the combined feedback variation with MedAlpaca and PMC-LLaMA achieves the best results. This suggests that even though the generalized reasoning experts might provide semantically appropriate information, they might not be as effective in maintaining contextual consistency for a particular domain as the fine-tuned experts. Figure 13 (a) further illustrates that the architecture for combined feedback from PMC-LLaMA and MedAlpaca is able to process and generate responses from medical information in a much more sound manner. Furthermore, when combined with a better consistency score (closer to zero), the improved ability of the model to deal with important tasks such as diagnosis and risk assessment is highlighted. A better Rouge-L and Consistency score in a single model is particularly important to make reliable medical conclusions that can directly impact the safety of everyone involved. Figure 13 (b) provides insights into how the approach utilizing PMC-LLaMA and MedAlpaca reinforces the ability of specially trained models to generate better and relevant responses by achieving a higher consistency score compared to other variations.

In Figure 12, the performances of various architectural variations are examined regarding their accuracy in determining the final decision label (Yes/No/Maybe) in PubMedQA questions. Although the proposed approach, which integrates feedback from both PMC-LLaMA and MedAlpaca, yielded slightly fewer correct decision labels (59) compared to the individual feedback (61 for PMC-LLaMA and 61 for MedAlpaca), it is pertinent to emphasize that this approach results in a reduction of incorrect decision labels to 9. This diminution in incorrect labels, coupled with the substantial number of inconclusive decision labels (32), underscores the model's capacity to minimize erroneous conclusions and highlights the necessity for a comprehensive understanding of the question prior to arriving at any definitive conclusion, which is critical for inference in medical tasks. As illustrated in Figure 14, the approach employing Ministral 8B and Claude 3-haiku demonstrates a higher number of correct conclusions (69) and fewer incorrect conclusions (16), indicating that generalized experts are proficient at effectively capturing reasoning patterns. Although the elevated number of correct answers reflects the model's confidence in identifying the appropriate

decision label, the lower evaluation scores (Rouge-L: 0.229 and Consistency: -2.110) imply that specific expertise, as exemplified by PMC-LLaMA and MedAlpaca, is more adept at providing insights on medical reasoning. Consequently, further research in this area could investigate the integration of a general LLM with a domain-specific fine-tuned LLM to augment the foundational LLM's capacity to generate semantically appropriate responses and maintain contextual consistency.

2) Results

The experiment and analysis conducted on the PubMedQA dataset to evaluate the proposed architecture's capability in handling domain-specific datasets have yielded several notable insights. The integration of the proposed approach with fine-tuned LLM experts, such as PMC-LLaMA and MedAlpaca, has resulted in a considerable reduction in the number of incorrect conclusions, along with improved evaluation scores. This outcome highlights the model's emphasis on rigorously generating precise and dependable conclusions. Future research, as outlined in the Future Research Section, could investigate the implementation of a hierarchical structure to assess the feedback of the experts in relation to specific tasks, thereby potentially enhancing the architecture's overall performance. A notable point of the experiment and analysis conducted on PubMedQA dataset is that queries with conclusion labeled as "maybe" have a higher number of incorrect final decisions compared to 'yes' or 'no' labeled queries. This can be attributed to the definite decision boundary that can be drawn with a 'yes' or 'no' label due to which a performance variance in inconclusive decisions has been noted across the subsets in the result section. Reliance on domain specific-experts helped. However, ablation studies can be further performed to reduce the ambiguity of 'maybe' labels.

VIII. PRACTICAL CONSIDERATIONS & FUTURE WORK

While the proposed architecture demonstrates strong performance in both general and domain-specific reasoning tasks, several practical limitations must be acknowledged. The iterative nature of the feedback loop introduces response latency and computational overhead mainly due to the involvement of complex queries and multi-round reasoning. Lightweight models like Ministral 8B helps in mitigation of such issues, but deeper models like Claude 3-haiku incurs higher memory and processing demands (refer to Figure 15). These issues can hinder real-time or large-scale inference, Figure 15 outlines the quantitative impact of these constraints along with possible solutions, such as model parallelization and asynchronous feedback mechanisms.

Additionally, the current setup allows the expert LLMs to function independently and incorporating a collaboration strategy between expert LLMs may enhance the system's ability to synthesize diverse reasoning strategies. In future iterations, incorporating expert consensus mechanisms, confidence-weighted feedback aggregation, or dynamic ex-

pert activation (as suggested in Figure 15) could help resolve these issues and improve feedback integration.

While this study relies on widely adopted benchmarks (HotPotQA, SimpleQA and PubMedQA) that have published validation protocols and are generally considered representative for multi-step reasoning tasks in QA, they are not immune to hidden biases. HotPotQA dataset, for instance, might lead the model to overly rely on supporting facts, SimpleQA dataset's emphasis on short and fact-seeking questions can limit the model to understand tasks that require broader inference strategies, and the use of the 'maybe' conclusion label in PubMedQA dataset can introduce uncertainty in evaluation since different models can interpret 'maybe' label inconsistently. To mitigate these hidden biases avenues for potential future works are mentioned in Figure 15.

Finally, the architecture has been effectively evaluated using question-answering benchmarks; however, future research that involves applying this architecture to real-world scenarios with dynamic environments and comprehensive toolsets is essential to evaluate the model's capacity to manage unforeseen outcomes and ambiguity in natural language reasoning. Moreover, the incorporation of human feedback offers a significant potential to augment the safety and reliability of decision-making processes, particularly in domains such as healthcare, software development, and financial services.

IX. CONCLUSION

The integration of an expert LLM layer enhances the precision and consistency of the generated responses by providing targeted feedback, thus enabling the base LLM to engage in an iterative self-reflection process over its previous reasoning paths. The evaluation results illustrate the architecture's robustness in reducing agent hallucination and inconsistent responses, phenomena commonly observed in multi-step reasoning contexts. Moreover, the study has explored the potential of the proposed system in addressing domain-specific tasks with PubMedQA, and its application could be further investigated in real-world interactive settings to develop multi-agent and brokered systems tailored for each domain. The current experimentation is confined to controlled environments, which can be expanded to real-world scenarios to further optimize the system's capability to handle substantial volumes and intricate queries during inference. This progression could potentially augment the system's interactivity. This research lays the groundwork for the development of more contextually aware agentic systems capable of generating appropriate responses by leveraging expert-driven insights for various critical applications.

A structured analysis of the system's practical challenges and potential improvements is summarized in Figure 15. These include computational overhead, latency, expert disagreement, scalability limitations, and the need for more robust multilingual and domain-adaptive capabilities. Future work will focus on addressing these through strategies like lightweight expert distillation, dynamic expert routing, asyn-

Challenge	Key Findings	Quantitative Impact	Proposed Solutions for Future Work
Computational Overhead	Multi-expert architecture increases resource requirements	- Increase in computational resources (upto 3X) - Higher GPU memory for expert models (upto 2X) - Total token processing increases from baseline to multi-expert configuration upto 300%	- Knowledge distillation to compress expert knowledge - Research on lightweight expert "adapter" layers - Dynamic expert activation based on task classification
Latency Concerns	Iterative feedback mechanism introduces processing delays	- Average latency increase per trial is around 4s - Upto 8s per trial for complex medical queries - Latency of upto 40s for full 5-trial cycle as compared to 3-5s latency for single-pass	- Model parallelization - Persistent expert instances - Asynchronous feedback mechanisms
Scalability Challenges	Performance degrades with very large or complex queries	Degradation observed when: - Contexts exceed 10,000 tokens - Reasoning requires >10 logical steps - Queries span >3 distinct domains - Diminishing returns after 7-8 cumulative experts	- Sparse expert activation - More efficient feedback encoding - Retrieval-augmented reflection
Expert Disagreement	Conflicting expert feedback creates challenges in resolution	- Contradictory guidance: In 18% of HotPotQA cases and 32% of PubMedQA cases - Impact: Decision paralysis (7% of cases), Inconsistent reasoning (11%), Excessive deference (15%)	- Dedicated arbiter model - Confidence-weighted feedback aggregation - Voting mechanisms for conflict resolution
Domain Generalization	Performance drops in unfamiliar domains	- Accuracy drops upto 30% in out-of-domain scientific literature - Limited ability to recognize knowledge boundaries	- Domain classification preprocessing - Uncertainty quantification mechanisms - Framework for dynamic expert incorporation
Dataset Biases	Current benchmarks may contain hidden biases affecting evaluation	- HotPotQA reliance on supporting facts - SimpleQA's emphasis on short queries/PubMedQA's ambiguity in "maybe" labels	- Cross-validation using repeated experiments - Adversarial question perturbations - Balanced dataset construction
Evaluation Metrics Limitations	Current metrics may not fully capture reasoning quality	- N-gram based metrics miss semantic similarity - Consistency scores don't fully measure reasoning coherence	- Incorporate BERTScore, MoverScore, and BLEU - Develop specialized metrics for multi-step reasoning
Multilingual Support	Current implementation limited to English datasets	- Performance degradation with non-English queries - Challenges with diverse sentence structures	- Incorporate language-specific datasets - Develop language-agnostic architectures
Hierarchical Expert Structure	Current flat expert layer may limit specialized feedback	- Suboptimal performance in highly specialized domains - Inefficient expert utilization	- Implement hierarchical structure of expert-level LLMs - Create domain-expert routing mechanisms
Ablation Studies	Limited understanding of individual component contributions	Uncertain impact of removing specific architectural elements	- Conduct comprehensive ablation studies - Test with/without base agent - Evaluate individual expert contributions
Human Feedback Integration	Current system lacks human expert integration	Limited ability to incorporate domain expert feedback	- Develop human-in-the-loop feedback mechanisms - Create interfaces for expert intervention

FIGURE 15. Comprehensive analysis of architectural challenges, performance bottlenecks, and proposed future directions for improving expert-augmented LLM frameworks. This table summarizes key limitations in the current system, their quantitative impact on reasoning tasks, and actionable strategies to enhance scalability, accuracy, and efficiency across diverse domains.

chronous feedback loops, and the integration of domain experts in a human-in-the-loop setting.

Overall, this study provides a compelling step towards building intelligent, feedback-aware systems that can adapt, refine, and reason with increasing sophistication across a wide variety of applications.

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APPENDIX

A. QUALITATIVE ANALYSIS

The integration of Claude 3-haiku and Ministral 8B into the proposed architecture offers a synergistic effect that significantly elevates HotpotQA's performance in multi-step reasoning tasks. Claude 3-haiku, with its deep reasoning capabilities, excels at providing nuanced insights and complex problem-solving strategies, which are crucial for understanding intricate query requirements. On the other hand, Ministral 8B brings unparalleled factual accuracy and robustness, ensuring that the generated responses are semantically consistent and grounded in reliable information. When combined within the iterative feedback mechanism, these two models work in tandem to refine each other's contributions. Claude 3-haiku's reasoning depth complements Ministral 8B's factual accuracy by offering diverse perspectives and detailed analyses that enhance the base model's understanding of the query context. Conversely, the base model provides structured responses that guide Claude and Ministral towards more relevant solutions. This collaborative interplay is visually depicted in Figure 8 (c), where the rate of correct answers steadily increases over four trials (from 34 to 48 percent). Quantitatively, it is observed that the error rate (percentage of incorrect conclusions) drops from 34 percent in Trial 1 to 12 percent by Trial 5 - a reduction of 22 percentage points - highlighting a significant impact of iterative feedback on model accuracy. In Trial 1 (Appendix B), Claude 3-haiku and Ministral 8B both employ slightly different strategies, though their objectives are aligned. Claude adopts a step-by-step approach, while Ministral 8B utilizes a more targeted strategy. However, Trial 1 is unsuccessful across 34 percent of the questions mainly due to the model being halted after finishing maximum passes on a question or mismatching the facts in the queries. In Trial 2 (Appendix B), Claude expands the search scope to gather broader context, enhancing the effectiveness of future searches. Ministral 8B adapts by employing clarifying search terms and alternative methods, demonstrating its flexibility in problem-solving. This adaptation leads to partial improvement and sets the stage for accuracy gains in subsequent iterations and also allow the model to enhance the reasoning steps as evidenced by the increased number of passes (6 passes) in this trial. In Trial 3 (Appendix B), Claude focuses on key information regarding McNatt's career and team details, while Ministral 8B refines its approach by incorporating synonyms and additional sources to ensure comprehensive coverage of the necessary data. By Trial 4 (Appendix B), their combined efforts—characterized by specificity, breadth, effective synonym use, and iterative refinement—ultimately yields the correct answer and reduces the number of passes required (from 6 passes in trial 2 to 3 passes in Trial 4). This is especially critical since the observations made by the model are also refined as the trials progresses (Observations made in trial 3 and trial 4 lead to different thoughts) and the base model draws context from the observations profoundly. These observations highlighted

areas for improvement but ultimately validated the effectiveness of their collaborative approach. While this approach offers significant improvements in both generation quality and model accuracy, there are areas where further exploration could enhance its effectiveness. For instance, incorporating dynamic weighting mechanisms between Claude's reasoning depth and Ministral's factual accuracy could optimize the balance of contributions based on query complexity. Additionally, exploring alternative feedback aggregation strategies might unlock new potential for refining multi-expert architectures. In conclusion, the proposed architecture represents a promising advancement in enhancing HotpotQA's reasoning capabilities through the integration of diverse expert models. By leveraging the unique strengths of Claude 3-haiku and Ministral 8B within an iterative feedback framework, the architecture not only improves response accuracy but also fosters continuous adaptation to complex reasoning tasks.

B. REFLEXION + REACT AGENT WITH MINISTRAL 8B + CLAUDE 3-HAIKU FEEDBACK ON A GENERALIZED REASONING TASK

Question: What team was led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat"?

1) Trial 1

Thought 1: I need to search for the two-time All-American basketball player nicknamed "Scat" and find out what team he led to victory in 1939.

Action 1: Search["Scat" basketball player]

Observation 1: James Carlos McNatt (December 19, 1918 – December 23, 2000) was an All-American basketball player for the Oklahoma Sooners and the AAU's Phillips 66ers. At Oklahoma, McNatt led his team to the first-ever NCAA final Four in 1939, and at Phillips 66, McNatt guided the 66ers (also called the "Oilers") to four consecutive AAU national championships (1943, 1944, 1945, and 1946). He was a two-time All-American at Oklahoma (1939, 1940) and a four-time AAU All-American for Phillips 66 (1943, 1944, 1945, 1946). The speedy player came to be known by his nickname "Scat" McNatt, a moniker originally traced back to the term "Boy Scats" which sportswriters had used to describe McNatt's fast-breaking, sophomore-led 1937-38 Oklahoma Sooners basketball team. McNatt grew up in Norman, Oklahoma, attended Norman High School, and then opted to stay in his hometown to play basketball for the University of Oklahoma. During his collegiate career at Oklahoma, McNatt broke the Big Six Conference single-game scoring record on two occasions. As a junior, he set the mark with 29 points against Iowa State University on February 18, 1939. Then as a senior, he broke his own record with 30 points against Nebraska on February 9, 1940. Another of Oklahoma's conference rivals in the Big Six was the University of Kansas, coached by the legendary Phog Allen. McNatt's Sooners teams were coached by Hugh McDermott and Bruce Drake and won back-to-back Big Six Conference championships in McNatt's junior (1938–39) and senior (1939–40) seasons.

McNatt was the conference's leading scorer during the 1939-40 season. In the 1939 NCAA Division I men's basketball tournament, McNatt led his team to the first-ever Final Four. He topped all scorers with 12 points in the national quarterfinals ("Western Regional semifinal") as the Sooners defeated Utah State 50-39. In the national semifinal ("Western Regional final") against the University of Oregon, McNatt was again the Sooners' leading scorer with 12 points as the Sooners fell to the eventual-champion Ducks. Both games were played in San Francisco at the 1939 Golden Gate International Exposition. To advance to the Western Regional, the Big Six Conference Champion Sooners first had to defeat Missouri Valley Conference Champion Oklahoma A&M, coached by Hank Iba and led by Jesse Renick, on a neutral court in Oklahoma City one week earlier. After graduating from Oklahoma, McNatt moved on to a career in Amateur Athletic Union basketball, an early basketball league that preceded the National Basketball Association. Though runners up in the 1942 AAU National Tournament with a team featuring McNatt and former Stanford star Hank Luisetti, Phillips 66 won the next four consecutive AAU National Championships, led by the play of McNatt and Gordon Carpenter, both of whom were selected to the AAU All-America teams during these four seasons. In 1943 and 1944, the 66ers defeated Denver's American Legion team 57-40 and then 50-43 in the tournament-final games. In the 1945 championship game, a 47-46 win over Denver Ambrose, McNatt led Phillips 66 in scoring with 15 points, outdueling Denver star Robert Gruenig. In McNatt's final season, Phillips 66 defeated the San Diego Dons by the score of 45-34 in the AAU championship game. McNatt was inducted into the Helms Athletic Foundation's Basketball Hall of Fame in 1960. After his basketball career, McNatt worked as a petroleum engineer for the Phillips Petroleum Company, and held numerous supervisory and management positions at Phillips for his entire career, retiring July 1, 1980.

Thought 2: The player nicknamed "Scat" is James Carlos McNatt, who led the Oklahoma Sooners to the first-ever NCAA Final Four in 1939. I need to find out which team they defeated in the national quarterfinals.

Action 2: Lookup[national quarterfinals]

Observation 2: (Result 1/1) James Carlos McNatt (December 19, 1918 – December 23, 2000) was an All-American basketball player for the Oklahoma Sooners and the AAU's Phillips 66ers. At Oklahoma, McNatt led his team to the first-ever NCAA final Four in 1939, and at Phillips 66, McNatt guided the 66ers (also called the "Oilers") to four consecutive AAU national championships (1943, 1944, 1945, and 1946). He was a two-time All-American at Oklahoma (1939, 1940) and a four-time AAU All-American for Phillips 66 (1943, 1944, 1945, 1946). The speedy player came to be known by his nickname "Scat" McNatt, a moniker originally traced back to the term "Boy Scats" which sportswriters had used to describe McNatt's fast-breaking, sophomore-led 1937-38 Oklahoma Sooners basketball team. McNatt grew up in Norman, Oklahoma, attended Norman High School, and then opted to stay in his hometown to play basketball for the University of

Oklahoma. During his collegiate career at Oklahoma, McNatt broke the Big Six Conference single-game scoring record on two occasions. As a junior, he set the mark with 29 points against Iowa State University on February 18, 1939. Then as a senior, he broke his own record with 30 points against Nebraska on February 9, 1940. Another of Oklahoma's conference rivals in the Big Six was the University of Kansas, coached by the legendary Phog Allen. McNatt's Sooners teams were coached by Hugh McDermott and Bruce Drake and won back-to-back Big Six Conference championships in McNatt's junior (1938-39) and senior (1939-40) seasons. McNatt was the conference's leading scorer during the 1939-40 season. In the 1939 NCAA Division I men's basketball tournament, McNatt led his team to the first-ever Final Four. He topped all scorers with 12 points in the national quarterfinals ("Western Regional semifinal") as the Sooners defeated Utah State 50-39. In the national semifinal ("Western Regional final") against the University of Oregon, McNatt was again the Sooners' leading scorer with 12 points as the Sooners fell to the eventual-champion Ducks. Both games were played in San Francisco at the 1939 Golden Gate International Exposition. To advance to the Western Regional, the Big Six Conference Champion Sooners first had to defeat Missouri Valley Conference Champion Oklahoma A&M, coached by Hank Iba and led by Jesse Renick, on a neutral court in Oklahoma City one week earlier. After graduating from Oklahoma, McNatt moved on to a career in Amateur Athletic Union basketball, an early basketball league that preceded the National Basketball Association. Though runners up in the 1942 AAU National Tournament with a team featuring McNatt and former Stanford star Hank Luisetti, Phillips 66 won the next four consecutive AAU National Championships, led by the play of McNatt and Gordon Carpenter, both of whom were selected to the AAU All-America teams during these four seasons. In 1943 and 1944, the 66ers defeated Denver's American Legion team 57-40 and then 50-43 in the tournament-final games. In the 1945 championship game, a 47-46 win over Denver Ambrose, McNatt led Phillips 66 in scoring with 15 points, outdueling Denver star Robert Gruenig. In McNatt's final season, Phillips 66 defeated the San Diego Dons by the score of 45-34 in the AAU championship game. McNatt was inducted into the Helms Athletic Foundation's Basketball Hall of Fame in 1960. After his basketball career, McNatt worked as a petroleum engineer for the Phillips Petroleum Company, and held numerous supervisory and management positions at Phillips for his entire career, retiring July 1, 1980.

Thought 3: In the national quarterfinals, the Oklahoma Sooners led by James Carlos McNatt defeated Utah State 50-39. I need to find out which team they faced in the national semifinal.

Action 3: Lookup[national semifinal]

Observation 3: (Result 1/1) James Carlos McNatt (December 19, 1918 – December 23, 2000) was an All-American basketball player for the Oklahoma Sooners and the AAU's Phillips 66ers. At Oklahoma, McNatt led his team to the first-ever NCAA final Four in 1939, and at Phillips 66, McNatt guided

the 66ers (also called the "Oilers") to four consecutive AAU national championships (1943, 1944, 1945, and 1946). He was a two-time All-American at Oklahoma (1939, 1940) and a four-time AAU All-American for Phillips 66 (1943, 1944, 1945, 1946). The speedy player came to be known by his nickname "Scat" McNatt, a moniker originally traced back to the term "Boy Scats" which sportswriters had used to describe McNatt's fast-breaking, sophomore-led 1937-38 Oklahoma Sooners basketball team. McNatt grew up in Norman, Oklahoma, attended Norman High School, and then opted to stay in his hometown to play basketball for the University of Oklahoma. During his collegiate career at Oklahoma, McNatt broke the Big Six Conference single-game scoring record on two occasions. As a junior, he set the mark with 29 points against Iowa State University on February 18, 1939. Then as a senior, he broke his own record with 30 points against Nebraska on February 9, 1940. Another of Oklahoma's conference rivals in the Big Six was the University of Kansas, coached by the legendary Phog Allen. McNatt's Sooners teams were coached by Hugh McDermott and Bruce Drake and won back-to-back Big Six Conference championships in McNatt's junior (1938-39) and senior (1939-40) seasons. McNatt was the conference's leading scorer during the 1939-40 season. In the 1939 NCAA Division I men's basketball tournament, McNatt led his team to the first-ever Final Four. He topped all scorers with 12 points in the national quarterfinals ("Western Regional semifinal") as the Sooners defeated Utah State 50-39. In the national semifinal ("Western Regional final") against the University of Oregon, McNatt was again the Sooners' leading scorer with 12 points as the Sooners fell to the eventual-champion Ducks. Both games were played in San Francisco at the 1939 Golden Gate International Exposition. To advance to the Western Regional, the Big Six Conference Champion Sooners first had to defeat Missouri Valley Conference Champion Oklahoma A&M, coached by Hank Iba and led by Jesse Renick, on a neutral court in Oklahoma City one week earlier. After graduating from Oklahoma, McNatt moved on to a career in Amateur Athletic Union basketball, an early basketball league that preceded the National Basketball Association. Though runners up in the 1942 AAU National Tournament with a team featuring McNatt and former Stanford star Hank Luisetti, Phillips 66 won the next four consecutive AAU National Championships, led by the play of McNatt and Gordon Carpenter, both of whom were selected to the AAU All-America teams during these four seasons. In 1943 and 1944, the 66ers defeated Denver's American Legion team 57-40 and then 50-43 in the tournament-final games. In the 1945 championship game, a 47-46 win over Denver Ambrose, McNatt led Phillips 66 in scoring with 15 points, outdueling Denver star Robert Gruenig. In McNatt's final season, Phillips 66 defeated the San Diego Dons by the score of 45-34 in the AAU championship game. McNatt was inducted into the Helms Athletic Foundation's Basketball Hall of Fame in 1960. After his basketball career, McNatt worked as a petroleum engineer for the Phillips Petroleum Company, and held numerous supervisory and management

positions at Phillips for his entire career, retiring July 1, 1980.
Correct answer: Oklahoma Sooners

2) Trial 2

Reflections: - Based on the previous reasoning trial, there are a few potential reasons for the failure to answer the question: 1. Insufficient information: The initial search for "Scat" basketball player provided a lot of relevant background information, but did not directly answer the question of what team was led to victory in 1939. More targeted searches were needed to find the specific team that the "Scat" player (James Carlos McNatt) led to victory that year. 2. Difficulty navigating search results: The agent seemed to have trouble finding the specific information about the 2004 film that Kam Heskin appeared in. The repeated searches for "The Prince & Me (2004 film)" did not yield the necessary information. To mitigate these issues, a revised high-level plan could be: 1. For the "Scat" basketball player question: - Start by searching for "James Carlos McNatt" to get more details on his basketball career. - Specifically look for information on the teams he played for and any championships or tournament appearances. - Once the 1939 NCAA Final Four appearance is identified, search for details on the specific
- ****Diagnosis:**** The failure in this trial was due to not directly searching for the team led to victory in 1939 by the player nicknamed "Scat." Instead, I focused on the player's career and achievements, which led to a lack of direct information about the team's victory. ****New Plan:**** To mitigate this failure, I will directly search for the team led to victory in 1939 by the player nicknamed "Scat" and then look up the specific details of their victory.

Thought 1: I need to search for the team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat."

Action 1: Search[1939 NCAA basketball championship Scat player victory]

Observation 1: Could not find [1939 NCAA basketball championship Scat player victory]. Similar: ['Al Scates', 'UCLA Bruins', 'UCLA-USC rivalry', 'Derrick Henry', 'University of California, Los Angeles', 'Willie "Woo Woo" Wong', 'List of Pawn Stars episodes', '1975 in baseball']

Thought 2: I need to refine my search keywords to directly find the team led to victory in 1939 by the player nicknamed "Scat."

Action 2: Search[1939 NCAA basketball championship winner Scat player]

Observation 2: Could not find [1939 NCAA basketball championship winner Scat player]. Similar: ['UCLA Bruins', 'List of University of California, Los Angeles people', 'UCLA-USC rivalry', 'Derrick Henry', 'University of California, Los Angeles', 'Willie "Woo Woo" Wong', 'List of Pawn Stars episodes', '1975 in baseball']

Thought 3: I need to focus on finding the specific team that the player nicknamed "Scat" led to victory in 1939. I should try searching for the player's name and the year of the victory together to get more targeted results.

Action 3: Search[James Carlos McNatt 1939 NCAA basketball championship victory]

Observation 3: Could not find [James Carlos McNatt 1939 NCAA basketball championship victory]. Similar: ['Bill Russell']

Thought 4: I need to try a different approach to find the team led to victory in 1939 by the player nicknamed "Scat." I should search for James Carlos McNatt's involvement in the 1939 NCAA basketball championship specifically.

Action 4: Search[James Carlos McNatt 1939 NCAA basketball championship team]

Observation 4: Could not find [James Carlos McNatt 1939 NCAA basketball championship team]. Similar: ['Jimmy McNatt', 'List of AAU men's basketball champions', 'Bill Russell', 'List of sportsperson-politicians']

Thought 5: I need to further refine my search to find the specific team that James Carlos McNatt led to victory in the 1939 NCAA basketball championship. I should try searching for the 1939 NCAA basketball championship winner and James Carlos McNatt's involvement.

Action 5: Search[1939 NCAA basketball championship winner James Carlos McNatt]

Observation 5: Could not find [1939 NCAA basketball championship winner James Carlos McNatt]. Similar: ["List of AAU men's basketball champions", 'Bill Russell', 'List of sportsperson-politicians']

Thought 6: I need to focus on finding the specific team that James Carlos McNatt led to victory in the 1939 NCAA basketball championship. I should try searching for the 1939 NCAA basketball championship winner and any information related to James Carlos McNatt's team.

Action 6: Search[1939 NCAA basketball championship winner James Carlos McNatt team]

Observation 6: Could not find [1939 NCAA basketball championship winner James Carlos McNatt team]. Similar: ["List of AAU men's basketball champions", 'Bill Russell', 'List of sportsperson-politicians']

Correct answer: Oklahoma Sooners

3) Trial 3

Reflections: - Based on the previous reasoning trial, there are a few potential reasons for the failure to answer the question: 1. Insufficient information: The initial search for "Scat" basketball player provided a lot of relevant background information, but did not directly answer the question of what team was led to victory in 1939. More targeted searches were needed to find the specific team that the "Scat" player (James Carlos McNatt) led to victory that year. 2. Difficulty navigating search results: The agent seemed to have trouble finding the specific information about the 2004 film that Kam Heskin appeared in. The repeated searches for "The Prince & Me (2004 film)" did not yield the necessary information. To mitigate these issues, a revised high-level plan could be: 1. For the "Scat" basketball player question: - Start by searching for "James Carlos McNatt" to get more details on his basketball career. - Specifically look for information on the

teams he played for and any championships or tournament appearances. - Once the 1939 NCAA Final Four appearance is identified, search for details on the specific

- ****Diagnosis:**** The failure in this trial was due to not directly searching for the team led to victory in 1939 by the player nicknamed "Scat." Instead, I focused on the player's career and achievements, which led to a lack of direct information about the team's victory. ****New Plan:**** To mitigate this failure, I will directly search for the team led to victory in 1939 by the player nicknamed "Scat" and then look up the specific details of their victory.

- Based on the previous reasoning trial, it seems the main reason for the failure to answer the question was the inability to find the specific information needed, even after several attempts at refining the search queries. A few possible reasons for this failure: 1. Lack of relevant information in the Docstore API: The Docstore API may not have contained the necessary information to answer this specific question about the 1939 NCAA basketball championship winner led by a player nicknamed "Scat". The search results did not provide any direct information about this event and player. 2. Difficulty in formulating the right search queries: The agent struggled to find the right combination of keywords to retrieve the relevant information. Searching for the player's name, the year, and the championship event did not yield the desired results. To mitigate this type of failure in the future, the agent could devise a new, more comprehensive plan: 1. Start with a broader search to understand the context and identify any relevant information about the 1939 NCAA basketball championship, players, and teams. 2. Gather information about notable players and teams from that era, and look for any references to a player nicknamed "Scat"

- ****Diagnosis:**** The failure in this trial was due to a lack of clarity in the search terms and a failure to consider alternative search strategies. The initial search terms were too broad and did not yield relevant results. ****New Plan:**** 1. ****Clarify the Search Terms:**** Start with a more specific search term, such as "1939 NCAA basketball championship winner James Carlos McNatt." 2. ****Use Alternative Search Strategies:**** If the initial search does not yield results, try searching for "1939 NCAA basketball championship team James Carlos McNatt" or "James Carlos McNatt 1939 NCAA basketball championship team." 3. ****Refine Search Based on Observations:**** If the search results are still not relevant, refine the search further based on the similar results provided. For example, if "Bill Russell" appears in the similar results, try searching for "Bill Russell 1939 NCAA basketball championship team."

Thought 1: I need to search for the team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat."

Action 1: Search[1939 NCAA basketball championship winner "Scat" player]

Observation 1: Derrick Lamar Henry Jr. (born January 4, 1994) is an American professional football running back for the Baltimore Ravens of the National Football League (NFL). Nicknamed "King Henry", he is known for his imposing style

of play and larger build than the average running back. Born and raised in Yulee, Florida, Henry set the national high school football record for career rushing yards with the Yulee Hornets. He played college football for the Alabama Crimson Tide, where during his junior season, he broke Herschel Walker's single-season college rushing yards record in the Southeastern Conference (SEC), won the 2015 Heisman Trophy, the Doak Walker Award, the Maxwell Award, and the Walter Camp Award, and was a key part of the 2015 Alabama Crimson Tide football team that won the 2016 College Football Playoff National Championship. Henry was selected by the Tennessee Titans in the second round of the 2016 NFL draft, where he was a four-time Pro Bowl selection. A backup for most of his first two seasons in the league, Henry emerged as the Titans' feature back near the end of the 2018 season. He led the NFL in rushing yards for the 2019 season, as well as rushing touchdowns (tied with Green Bay Packers running back Aaron Jones). In 2020, Henry again led the league in rushing yards and touchdowns and became the eighth player in NFL history to rush for over 2,000 yards in a single season, and the second Tennessee Titan to do so, after Chris Johnson achieved it in 2009. Henry won the 2020 Associated Press NFL Offensive Player of the Year Award. After missing more than half of the 2021 season to injury, he made his third and fourth Pro Bowl appearances in the subsequent two seasons. After eight seasons with Tennessee, Henry signed with the Ravens in 2024.

Thought 2: The search results did not provide relevant information about the team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat." I should refine my search terms to directly focus on the specific player and year.

Action 2: Search[1939 NCAA basketball championship winner "Scat" player team]

Observation 2: Derrick Lamar Henry Jr. (born January 4, 1994) is an American professional football running back for the Baltimore Ravens of the National Football League (NFL). Nicknamed "King Henry", he is known for his imposing style of play and larger build than the average running back. Born and raised in Yulee, Florida, Henry set the national high school football record for career rushing yards with the Yulee Hornets. He played college football for the Alabama Crimson Tide, where during his junior season, he broke Herschel Walker's single-season college rushing yards record in the Southeastern Conference (SEC), won the 2015 Heisman Trophy, the Doak Walker Award, the Maxwell Award, and the Walter Camp Award, and was a key part of the 2015 Alabama Crimson Tide football team that won the 2016 College Football Playoff National Championship. Henry was selected by the Tennessee Titans in the second round of the 2016 NFL draft, where he was a four-time Pro Bowl selection. A backup for most of his first two seasons in the league, Henry emerged as the Titans' feature back near the end of the 2018 season. He led the NFL in rushing yards for the 2019 season, as well as rushing touchdowns (tied with Green Bay Packers running back Aaron Jones). In 2020, Henry again led the league in

rushing yards and touchdowns and became the eighth player in NFL history to rush for over 2,000 yards in a single season, and the second Tennessee Titan to do so, after Chris Johnson achieved it in 2009. Henry won the 2020 Associated Press NFL Offensive Player of the Year Award. After missing more than half of the 2021 season to injury, he made his third and fourth Pro Bowl appearances in the subsequent two seasons. After eight seasons with Tennessee, Henry signed with the Ravens in 2024.

Thought 3: The search results are still not relevant to the question. I should try a different approach to find the team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat."

Action 3: Search[1939 NCAA basketball championship winner "Scat" player team victory]

Observation 3: Derrick Lamar Henry Jr. (born January 4, 1994) is an American professional football running back for the Baltimore Ravens of the National Football League (NFL). Nicknamed "King Henry", he is known for his imposing style of play and larger build than the average running back. Born and raised in Yulee, Florida, Henry set the national high school football record for career rushing yards with the Yulee Hornets. He played college football for the Alabama Crimson Tide, where during his junior season, he broke Herschel Walker's single-season college rushing yards record in the Southeastern Conference (SEC), won the 2015 Heisman Trophy, the Doak Walker Award, the Maxwell Award, and the Walter Camp Award, and was a key part of the 2015 Alabama Crimson Tide football team that won the 2016 College Football Playoff National Championship. Henry was selected by the Tennessee Titans in the second round of the 2016 NFL draft, where he was a four-time Pro Bowl selection. A backup for most of his first two seasons in the league, Henry emerged as the Titans' feature back near the end of the 2018 season. He led the NFL in rushing yards for the 2019 season, as well as rushing touchdowns (tied with Green Bay Packers running back Aaron Jones). In 2020, Henry again led the league in rushing yards and touchdowns and became the eighth player in NFL history to rush for over 2,000 yards in a single season, and the second Tennessee Titan to do so, after Chris Johnson achieved it in 2009. Henry won the 2020 Associated Press NFL Offensive Player of the Year Award. After missing more than half of the 2021 season to injury, he made his third and fourth Pro Bowl appearances in the subsequent two seasons. After eight seasons with Tennessee, Henry signed with the Ravens in 2024.

Thought 4: The search results are still not providing the information needed to answer the question about the team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat." I should try a different approach to find the specific details.

Action 4: Search[1939 NCAA basketball championship winner "Scat" player team victory details]

Observation 4: Pawn Stars is an American reality television series that premiered on History on July 19, 2009. The series is filmed in Las Vegas, Nevada, where it chronicles the activi-

ties at the World Famous Gold & Silver Pawn Shop, a 24-hour family business operated by patriarch Richard "Old Man" Harrison, his son Rick Harrison, Rick's son Corey "Big Hoss" Harrison, and Corey's childhood friend, Austin "Chumlee" Russell. The descriptions of the items listed in this article reflect those given by their sellers and staff in the episodes, prior to their appraisal by experts as to their authenticity, unless otherwise noted.

Correct answer: Oklahoma Sooners

4) Trial 4

Reflections: - Based on the previous reasoning trial, there are a few potential reasons for the failure to answer the question: 1. Insufficient information: The initial search for "Scat" basketball player provided a lot of relevant background information, but did not directly answer the question of what team was led to victory in 1939. More targeted searches were needed to find the specific team that the "Scat" player (James Carlos McNatt) led to victory that year. 2. Difficulty navigating search results: The agent seemed to have trouble finding the specific information about the 2004 film that Kam Heskin appeared in. The repeated searches for "The Prince & Me (2004 film)" did not yield the necessary information. To mitigate these issues, a revised high-level plan could be: 1. For the "Scat" basketball player question: - Start by searching for "James Carlos McNatt" to get more details on his basketball career. - Specifically look for information on the teams he played for and any championships or tournament appearances. - Once the 1939 NCAA Final Four appearance is identified, search for details on the specific

- ****Diagnosis:**** The failure in this trial was due to not directly searching for the team led to victory in 1939 by the player nicknamed "Scat." Instead, I focused on the player's career and achievements, which led to a lack of direct information about the team's victory. ****New Plan:**** To mitigate this failure, I will directly search for the team led to victory in 1939 by the player nicknamed "Scat" and then look up the specific details of their victory.

- Based on the previous reasoning trial, it seems the main reason for the failure to answer the question was the inability to find the specific information needed, even after several attempts at refining the search queries. A few possible reasons for this failure: 1. Lack of relevant information in the Docstore API: The Docstore API may not have contained the necessary information to answer this specific question about the 1939 NCAA basketball championship winner led by a player nicknamed "Scat". The search results did not provide any direct information about this event and player. 2. Difficulty in formulating the right search queries: The agent struggled to find the right combination of keywords to retrieve the relevant information. Searching for the player's name, the year, and the championship event did not yield the desired results. To mitigate this type of failure in the future, the agent could devise a new, more comprehensive plan: 1. Start with a broader search to understand the context and identify any relevant information about the 1939 NCAA basketball championship, players,

and teams. 2. Gather information about notable players and teams from that era, and look for any references to a player nicknamed "Scat"

- ****Diagnosis:**** The failure in this trial was due to a lack of clarity in the search terms and a failure to consider alternative search strategies. The initial search terms were too broad and did not yield relevant results. ****New Plan:**** 1. ****Clarify the Search Terms:**** Start with a more specific search term, such as "1939 NCAA basketball championship winner James Carlos McNatt." 2. ****Use Alternative Search Strategies:**** If the initial search does not yield results, try searching for "1939 NCAA basketball championship team James Carlos McNatt" or "James Carlos McNatt 1939 NCAA basketball championship team." 3. ****Refine Search Based on Observations:**** If the search results are still not relevant, refine the search further based on the similar results provided. For example, if "Bill Russell" appears in the similar results, try searching for "Bill Russell 1939 NCAA basketball championship team."

- Based on the previous reasoning trial, it seems the main reason for the failure to answer the question was the inability to find the relevant information needed to identify the specific team led to victory in 1939 by a two-time All-American basketball player nicknamed "Scat". The search results provided information about unrelated topics, such as a football player named Derrick Henry, rather than the basketball player and team in question. To mitigate this failure, a new, concise, high-level plan could be: 1. Clearly identify the key pieces of information needed to answer the question - the name of the two-time All-American basketball player nicknamed "Scat" and the team they led to victory in 1939. 2. Perform a targeted search for the name of the basketball player, using the nickname "Scat" as a key identifier. This should help narrow down the results to the specific player in question. 3. Once the player is identified, search for information on the team they played for and any details about their 1939 championship victory. 4. Synthesize the information found in the search results to provide a concise answer to the

- ****Diagnosis:**** The failure in this trial was due to the incorrect search terms and the lack of a clear strategy to find the specific information needed to answer the question. ****New Plan:**** 1. ****Refine Search Terms:**** Start with a more specific search term, such as "1939 NCAA basketball championship winner 'Scat' player team." 2. ****Use Synonyms:**** If the initial search does not yield results, try using synonyms for "Scat" like "nickname" or "alias." 3. ****Check Related Searches:**** If the search results are still not relevant, check the "Related Searches" section for alternative terms or phrases. 4. ****Iterate and Refine:**** If the search results are still not providing the needed information, iterate and refine the search terms based on the information available in the search results.

Thought 1: I need to search for the two-time All-American basketball player nicknamed "Scat" and find out which team he led to victory in 1939.

Action 1: Search[James Carlos McNatt]

Observation 1: Could not find [James Carlos McNatt]. Sim-

ilar: ['Jimmy McNatt', 'List of AAU men's basketball champions', '2009–10 NBA season', 'List of songs about Paris', '2004–05 Cleveland Cavaliers season', 'List of NCAA Division I men's basketball career scoring leaders', 'List of Armchair Theatre episodes', 'Lucky Charms', 'List of NBA All-Stars', '1957 in literature']

Thought 2: The search for James Carlos McNatt did not yield relevant results. I should try searching for "Scat" basketball player instead to find the specific information about the team he led to victory in 1939.

Action 2: Search["Scat" basketball player]

Observation 2: James Carlos McNatt (December 19, 1918 – December 23, 2000) was an All-American basketball player for the Oklahoma Sooners and the AAU's Phillips 66ers. At Oklahoma, McNatt led his team to the first-ever NCAA final Four in 1939, and at Phillips 66, McNatt guided the 66ers (also called the "Oilers") to four consecutive AAU national championships (1943, 1944, 1945, and 1946). He was a two-time All-American at Oklahoma (1939, 1940) and a four-time AAU All-American for Phillips 66 (1943, 1944, 1945, 1946). The speedy player came to be known by his nickname "Scat" McNatt, a moniker originally traced back to the term "Boy Scats" which sportswriters had used to describe McNatt's fast-breaking, sophomore-led 1937-38 Oklahoma Sooners basketball team. McNatt grew up in Norman, Oklahoma, attended Norman High School, and then opted to stay in his hometown to play basketball for the University of Oklahoma. During his collegiate career at Oklahoma, McNatt broke the Big Six Conference single-game scoring record on two occasions. As a junior, he set the mark with 29 points against Iowa State University on February 18, 1939. Then as a senior, he broke his own record with 30 points against Nebraska on February 9, 1940. Another of Oklahoma's conference rivals in the Big Six was the University of Kansas, coached by the legendary Phog Allen. McNatt's Sooners teams were coached by Hugh McDermott and Bruce Drake and won back-to-back Big Six Conference championships in McNatt's junior (1938–39) and senior (1939–40) seasons. McNatt was the conference's leading scorer during the 1939-40 season. In the 1939 NCAA Division I men's basketball tournament, McNatt led his team to the first-ever Final Four. He topped all scorers with 12 points in the national quarterfinals ("Western Regional semifinal") as the Sooners defeated Utah State 50-39. In the national semifinal ("Western Regional final") against the University of Oregon, McNatt was again the Sooners' leading scorer with 12 points as the Sooners fell to the eventual-champion Ducks. Both games were played in San Francisco at the 1939 Golden Gate International Exposition. To advance to the Western Regional, the Big Six Conference Champion Sooners first had to defeat Missouri Valley Conference Champion Oklahoma A&M, coached by Hank Iba and led by Jesse Renick, on a neutral court in Oklahoma City one week earlier. After graduating from Oklahoma, McNatt moved on to a career in Amateur Athletic Union basketball, an early basketball league that preceded the National Basketball Association. Though runners up in the 1942 AAU

National Tournament with a team featuring McNatt and former Stanford star Hank Luisetti, Phillips 66 won the next four consecutive AAU National Championships, led by the play of McNatt and Gordon Carpenter, both of whom were selected to the AAU All-America teams during these four seasons. In 1943 and 1944, the 66ers defeated Denver's American Legion team 57-40 and then 50-43 in the tournament-final games. In the 1945 championship game, a 47-46 win over Denver Ambrose, McNatt led Phillips 66 in scoring with 15 points, outdueling Denver star Robert Gruenig. In McNatt's final season, Phillips 66 defeated the San Diego Dons by the score of 45-34 in the AAU championship game. McNatt was inducted into the Helms Athletic Foundation's Basketball Hall of Fame in 1960. After his basketball career, McNatt worked as a petroleum engineer for the Phillips Petroleum Company, and held numerous supervisory and management positions at Phillips for his entire career, retiring July 1, 1980.

Thought 3: The search for the "Scat" basketball player, James Carlos McNatt, provided the necessary information. He led the Oklahoma Sooners to the first-ever NCAA Final Four in 1939. Therefore, the team led to victory in 1939 by James Carlos McNatt, the "Scat" player, was the Oklahoma Sooners.

Action 3: Finish[Oklahoma Sooners]

Observation 3: Answer is CORRECT

Correct answer: Oklahoma Sooners

C. REFLEXION + COT AGENT WITH MINISTRAL 8B + CLAUDE 3-HAIKU FEEDBACK ON A MEDICAL REASONING TASK

Relevant Context: Specialty pharmaceuticals have evolved beyond their status as niche drugs designed to treat rare conditions and are now poised to become the standard of care in a wide variety of common chronic illnesses. Due in part to the cost of these therapies, payers are increasingly demanding evidence of their value. Determining the value of these medications is hampered by a lack of robust pharmacoeconomic data. To outline emerging strategies and case study examples for the medical and pharmacy benefits management of specialty pharmaceuticals. The promise of specialty pharmaceuticals: increased life expectancy, improved quality of life, enhanced workplace productivity, decreased burden of disease, and reduced health care spending comes at a significant cost. These agents require special handling, administration, patient education, clinical support, and risk mitigation. Additionally, specialty drugs require distribution systems that ensure appropriate patient selection and data collection. With the specialty pharmaceutical pipeline overflowing with new medicines and an aging population increasingly relying on these novel treatments to treat common diseases, the challenge of managing the costs associated with these agents can be daunting. Aided by sophisticated pharmacoeconomic models to assess value, the cost impacts of these specialty drugs can be appropriately controlled.

Question: The promise of specialty pharmaceuticals: are they worth the price?

1) Trial 1

Thought: The question is asking about the value of specialty pharmaceuticals in relation to their cost. Based on the context provided, specialty pharmaceuticals offer increased life expectancy, improved quality of life, enhanced workplace productivity, decreased burden of disease, and reduced health care spending. However, they come at a significant cost and require special handling, administration, and support. The challenge lies in managing these costs while ensuring the benefits are realized.

Action: Finish[Specialty pharmaceuticals offer significant benefits such as increased life expectancy, improved quality of life, enhanced workplace productivity, decreased burden of disease, and reduced health care spending. However, they come at a significant cost and require special handling, administration, and support. The challenge lies in managing these costs while ensuring the benefits are realized.]

Note: The Model didn't provide any decision label

Observation: Answer is INCORRECT

Correct answer: Current evidence suggests that when used in targeted patient populations, specialty pharmaceuticals may represent a good health care value.

Reflections: - Based on the feedback that my previous answer was incorrect, I believe the reason for the failure may be that my response was too long and detailed, rather than providing a more concise, high-level answer to the question. The question is asking whether the promise of specialty pharmaceuticals is worth the price. While I provided a comprehensive overview of the benefits and challenges associated with these drugs, I did not directly address whether the benefits outweigh the costs. To mitigate this failure, my new plan is to provide a more succinct, high-level response that directly answers the question. I will aim to weigh the pros and cons of specialty pharmaceuticals and provide a clear assessment of whether the benefits are worth the price.

New plan: The promise of specialty pharmaceuticals, such as increased life expectancy and improved quality of life, must be carefully weighed against their significant cost. While these drugs offer substantial benefits, their high price and the need for specialized handling and support can be a significant burden on the healthcare system. Ultimately, the value of specialty pharmaceuticals depends on the ability to manage their costs through robust pharmacoeconomic analysis and ensure that the benefits are realized for patients.

- The phrasing discrepancy in my previous response was due to the length and verbosity of the answer, which may not have been concise enough for the grading environment. In the future, I will aim to provide a more concise and direct answer. For example, I could have simply stated: "Yes, specialty pharmaceuticals are worth the price due to their significant benefits, but managing their costs is a challenge."

2) Trial 2

Thought: The question is asking whether the benefits of specialty pharmaceuticals outweigh the costs associated with

them. Given the context provided, which highlights the increased life expectancy, improved quality of life, and other benefits of these drugs, as well as the challenges related to their cost and management, I can conclude that specialty pharmaceuticals are indeed worth the price.

Action: Finish[Yes, specialty pharmaceuticals are worth the price due to their significant benefits, such as increased life expectancy, improved quality of life, and enhanced workplace productivity. Despite the challenges related to their cost and management, the value they provide in terms of patient outcomes and overall healthcare spending justifies their price.]

Note: The model provides the correct decision label 'Yes' along with an improved response

Correct answer: Current evidence suggests that when used in targeted patient populations, specialty pharmaceuticals may represent a good health care value.

D. REFLEXION + COT AGENT WITH MEDALPACA + PMC LLAMA FEEDBACK ON A MEDICAL REASONING TASK

Relevant Context: This study examines whether having a regular clinician for preventive care is associated with quality of care for young children, as measured by interpersonal quality ratings and content of anticipatory guidance. The National Survey of Early Childhood Health (NSECH), a nationally representative parent survey of health care quality for 2068 young US children fielded by the National Center for Health Statistics (NCHS). Bivariate and multivariate analyses evaluate associations between having a regular clinician for well child care and interpersonal quality, the content of anticipatory guidance, and timely access to care. In bivariate analysis, parents of children with a regular clinician for preventive care reported slightly higher interpersonal quality (69 vs. 65 on a 0-100 scale, $P = 0.01$). Content of anticipatory guidance received was slightly greater for children with a regular clinician (82 vs. 80 on a 0-100 scale, $P = 0.03$). In bivariate analysis, a regular clinician was associated with interpersonal quality only among African American and Hispanic children. In multivariate analyses, controlling for factors that could independently influence self-reports of experiences with care, interpersonal quality but not anticipatory guidance content was higher for children with a regular clinician.

Question: Does having a regular primary care clinician improve the quality of preventive care for young children?

1) Trial 1

Thought: The context provides information on the association between having a regular primary care clinician and the quality of preventive care for young children. While having a regular clinician is linked to higher interpersonal quality, the impact on anticipatory guidance content is more mixed. The multivariate analysis showed that only the association with interpersonal quality remained significant. Therefore, having a regular primary care clinician does improve the quality of preventive care for young children, particularly in terms of interpersonal quality.

Action: Finish[Having a regular primary care clinician improves the quality of preventive care for young children, particularly in terms of interpersonal quality.]

Observation: Answer is INCORRECT

Note: The model provides no decision label and an incomplete conclusion for the current question.

Correct answer: Having a regular primary care clinician is embraced in pediatrics, although team care among physicians is also widely practiced. For young children, having a regular clinician is associated with modest gains in interpersonal quality and no differences in content of anticipatory guidance. The benefit of having a regular clinician may primarily occur in interpersonal quality for subgroups of young children.

Reflections: - Based on the provided context and the question, it seems that my previous answer was not fully accurate. The key issue appears to be that while the context indicates having a regular clinician is associated with higher interpersonal quality, the evidence on the content of anticipatory guidance is more mixed. A possible reason for the failure could be that I did not fully capture the nuance in the findings, specifically the distinction between the impact on interpersonal quality versus the content of anticipatory guidance. The context suggests that the association with interpersonal quality was more clear-cut, while the impact on anticipatory guidance content was more modest. To mitigate this type of failure in the future, I could devise a new, more concise plan that focuses on: 1. Carefully parsing the details in the context, particularly any distinctions or nuances in the findings. 2. Ensuring my answer directly addresses the specific question being asked, rather than providing a more general summary. 3. Being more precise in my wording to accurately reflect the evidence presented, rather than making broad statements. For this specific question, a revised, more accurate answer could be: "The evidence suggests that having a regular primary care - The phrasing discrepancy in my answer was due to the verbosity and specificity of the response. The question was asking for a simple yes or no answer, and my response was too detailed. In the future, I will provide a more concise answer, such as "Yes, having a regular primary care clinician improves interpersonal quality but not anticipatory guidance content for young children."

- Based on the provided context, it seems that the reason my previous answer was incorrect is that I did not fully capture the nuance in the findings. The context indicates that having a regular primary care clinician is associated with: 1. Slightly higher interpersonal quality ratings (69 vs. 65 on a 0-100 scale, $p=0.01$) 2. Slightly greater content of anticipatory guidance received (82 vs. 80 on a 0-100 scale, $p=0.03$) However, the multivariate analysis showed that only the association with interpersonal quality remained significant after controlling for other factors, while the association with anticipatory guidance content did not. To improve my answer and better reflect the findings, I should have stated that having a regular primary care clinician is associated with improved interpersonal quality of preventive care for young children, but the impact on the content of anticipatory guidance is

more mixed. A more concise, high-level plan to address this type of question in the future would be: 1. Carefully read and extract the key findings from the provided context, noting any nuances or mixed results.

- The phrasing discrepancy in my answer was due to the inclusion of additional details that were not explicitly asked for in the question. The question only asked if having a regular primary care clinician improves quality of preventive care for young children, not the specific aspects of quality. In the future, I will focus on providing a concise answer that directly addresses the question without adding unnecessary details.

2) Trial 2

Thought: The context provides a detailed analysis of the association between having a regular clinician for preventive care and the quality of care for young children. While having a regular clinician is associated with higher interpersonal quality, the evidence is mixed regarding the content of anticipatory guidance. The multivariate analysis showed that interpersonal quality was higher for children with a regular clinician, but there was no significant improvement in anticipatory guidance content.

Action: Finish[Yes, Having a regular primary care clinician improves interpersonal quality but does not significantly improve the content of anticipatory guidance for young children.]

Note: The model provides a correct decision label 'Yes' along with a more complete explanation than the previous trial.

Correct answer: Having a regular primary care clinician is embraced in pediatrics, although team care among physicians is also widely practiced. For young children, having a regular clinician is associated with modest gains in interpersonal quality and no differences in content of anticipatory guidance. The benefit of having a regular clinician may primarily occur in interpersonal quality for subgroups of young children.

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